Answer key - Easy

Each answer is the letter replacing ?. Mark the questions before reading the deduction paths.

No.	Ans.	No.	Ans.	No.	Ans.	No.	Ans.	No.	Ans.
1	C	15	A	29	A	43	E	57	C
2	E	16	A	30	D	44	D	58	E
3	C	17	D	31	C	45	D	59	B
4	A	18	B	32	C	46	B	60	A
5	B	19	C	33	D	47	B	61	E
6	A	20	E	34	A	48	D	62	D
7	D	21	E	35	E	49	B	63	C
8	C	22	D	36	A	50	C	64	C
9	E	23	A	37	D	51	E	65	B
10	E	24	C	38	D	52	B	66	A
11	B	25	A	39	D	53	E	67	C
12	E	26	D	40	B	54	C	68	B
13	B	27	A	41	E	55	C	69	D
14	B	28	B	42	A	56	A	70	E

Verified answers

Every puzzle admits exactly one valid full grid under the A-E row and column rules. Both independent completion checks produced that same grid and the same marked-cell answer.

If your answer differs, follow the worked path from the original clues. Check the first step you missed rather than memorising the final letter.

Answer key - Moderate

Each answer is the letter replacing ?. Mark the questions before reading the deduction paths.

No.	Ans.	No.	Ans.	No.	Ans.	No.	Ans.	No.	Ans.
71	D	85	B	99	C	113	C	127	B
72	E	86	E	100	D	114	C	128	B
73	D	87	E	101	C	115	D	129	A
74	A	88	B	102	E	116	C	130	D
75	E	89	E	103	C	117	D	131	D
76	E	90	D	104	A	118	A	132	E
77	C	91	A	105	B	119	E	133	A
78	C	92	C	106	D	120	C	134	B
79	E	93	C	107	D	121	D	135	D
80	D	94	A	108	B	122	E	136	B
81	B	95	A	109	A	123	B	137	B
82	A	96	A	110	A	124	B	138	C
83	C	97	B	111	E	125	A	139	A
84	D	98	E	112	C	126	E	140	B

Verified answers

Every puzzle admits exactly one valid full grid under the A-E row and column rules. Both independent completion checks produced that same grid and the same marked-cell answer.

If your answer differs, follow the worked path from the original clues. Check the first step you missed rather than memorising the final letter.

Answer key - High

Each answer is the letter replacing ?. Mark the questions before reading the deduction paths.

No.	Ans.	No.	Ans.	No.	Ans.	No.	Ans.	No.	Ans.
141	D	155	D	169	B	183	E	197	A
142	E	156	C	170	D	184	A	198	E
143	B	157	E	171	B	185	C	199	D
144	B	158	C	172	D	186	A	200	D
145	A	159	E	173	E	187	E	201	C
146	D	160	A	174	C	188	A	202	A
147	C	161	C	175	E	189	D	203	B
148	B	162	A	176	C	190	E	204	E
149	E	163	C	177	B	191	A	205	C
150	C	164	D	178	B	192	E	206	A
151	D	165	C	179	B	193	B	207	B
152	B	166	B	180	D	194	D	208	C
153	A	167	E	181	A	195	D	209	A
154	C	168	B	182	D	196	A	210	E

Verified answers

Every puzzle admits exactly one valid full grid under the A-E row and column rules. Both independent completion checks produced that same grid and the same marked-cell answer.

If your answer differs, follow the worked path from the original clues. Check the first step you missed rather than memorising the final letter.

Worked solutions - Easy

Follow the numbered deductions. Magenta letters show the route to the shaded target. Other blank cells need not be filled.

Exercise 1 | Answer: C

Easy • 1 step in this route

	α	β	γ	δ	ε
1		C	B	D	A
2	A	B	D		
3	B	E	A	**C**	
4	D				E
5		D	E		B

1. δ3 = C: row 3 allows {C, D}; column δ allows {A, B, C, E}.

Exercise 2 | Answer: E

Easy • 1 step in this route

	α	β	γ	δ	ε
1	C	D	E		
2	A	C	B	E	
3		A		B	C
4	B	**E**		D	A
5	D		A		

1. β4 = E: row 4 allows {C, E}; column β allows {B, E}.

Exercise 3 | Answer: C

Easy • 1 step in this route

	α	β	γ	δ	ε
1	B	D		E	A
2	A		D	C	B
3	E	B	A		
4		**C**	B		
5	C	A	E		

1. β4 = C: in column β, C fits only in row 4.

Exercise 4 | Answer: A

Easy • 1 step in this route

	α	β	γ	δ	ε
1	E	B	C	D	A
2	A		E	B	
3		C			B
4		E	B	**A**	D
5	B				E

1. δ4 = A: in row 4, A fits only in column δ.

Worked solutions - Easy

Exercise 5 | Answer: B

Easy • 1 step in this route

	α	β	γ	δ	ε
1		E	D	B	C
2	E	D		A	B
3	C	A	B	E	D
4					
5	D	**B**			E

1. β5 = B: in row 5, B fits only in column β.

Exercise 6 | Answer: A

Easy • 1 step in this route

	α	β	γ	δ	ε
1	D	E		A	C
2	A	D		C	B
3	B		**A**		
4		B	D		
5	E		C	B	D

1. γ3 = A: in column γ, A fits only in row 3.

Exercise 7 | Answer: D

Easy • 1 step in this route

	α	β	γ	δ	ε
1		A	D		
2	A		E	B	C
3	D		B	E	
4	B				**D**
5	C	B	A	D	E

1. ε4 = D: in column ε, D fits only in row 4.

Exercise 8 | Answer: C

Easy • 1 step in this route

	α	β	γ	δ	ε
1	C		D	A	
2	D	A	E	B	
3		D	A		E
4		B	**C**	E	D
5	E	C			

1. γ4 = C: row 4 allows {A, C}; column γ allows {B, C}.

Worked solutions - Easy

Exercise 9 | Answer: E

Easy • 1 step in this route

	α	β	γ	δ	ε
1			D	A	B
2	E		A	B	
3	D	A		**E**	
4		B	C	D	E
5		D		C	A

1. δ3 = E: row 3 allows {B, C, E}; column δ allows {E}.

Exercise 10 | Answer: E

Easy • 1 step in this route

	α	β	γ	δ	ε
1	C			D	E
2	D	A	E	B	
3	A				B
4	**E**		B		D
5	B	E		C	

1. α4 = E: row 4 allows {A, C, E}; column α allows {E}.

Exercise 11 | Answer: B

Easy • 1 step in this route

	α	β	γ	δ	ε
1			A	E	
2	A	C		B	E
3	C			D	A
4	**B**		E	C	D
5			C		B

1. α4 = B: row 4 allows {A, B}; column α allows {B, D, E}.

Exercise 12 | Answer: E

Easy • 1 step in this route

	α	β	γ	δ	ε
1			B	C	E
2		E	C		
3		B		E	D
4		C	D	A	B
5		A	**E**	B	

1. γ5 = E: row 5 allows {C, D, E}; column γ allows {A, E}.

Worked solutions - Easy

Exercise 13 | Answer: B

Easy • 1 step in this route

	α	β	γ	δ	ε
1	**B**				C
2	E	B			
3	D	C	A		E
4		D	E		B
5		A	B	E	D

1. α1 = B: in column α, B fits only in row 1.

Exercise 14 | Answer: B

Easy • 1 step in this route

	α	β	γ	δ	ε
1		B	E	A	
2	E	A		C	D
3	**B**			E	A
4	C		A	D	
5			C	B	

1. α3 = B: in column α, B fits only in row 3.

Exercise 15 | Answer: A

Easy • 1 step in this route

	α	β	γ	δ	ε
1			C		A
2	C		E	**A**	D
3	A	D	B	E	
4	E				
5		C	A	B	E

1. δ2 = A: row 2 allows {A, B}; column δ allows {A, C, D}.

Exercise 16 | Answer: A

Easy • 1 step in this route

	α	β	γ	δ	ε
1	E	D		B	A
2	C		**A**		
3		C		D	E
4	B	E		A	C
5		A			B

1. γ2 = A: in row 2, A fits only in column γ.

Worked solutions - Easy

Exercise 17 | Answer: D

Easy • 1 step in this route

	α	β	γ	δ	ε
1			C	E	
2	B	A	E	C	
3	C		A	B	
4	A		B	**D**	
5	E	C		A	

1. δ4 = D: row 4 allows {C, D, E}; column δ allows {D}.

Exercise 18 | Answer: B

Easy • 1 step in this route

	α	β	γ	δ	ε
1			C	**B**	
2	A		E		
3	C		B	E	
4	E		D	A	C
5	B	C		D	

1. δ1 = B: row 1 allows {A, B, D, E}; column δ allows {B, C}.

Exercise 19 | Answer: C

Easy • 1 step in this route

	α	β	γ	δ	ε
1		**C**	D	B	E
2	B	A		E	D
3		B		A	
4	E				
5	C	E			A

1. β1 = C: row 1 allows {A, C}; column β allows {C, D}.

Exercise 20 | Answer: E

Easy • 1 step in this route

	α	β	γ	δ	ε
1			C	**E**	
2	C	A		B	E
3	B	E			
4		D	B	A	C
5			E	D	

1. δ1 = E: row 1 allows {A, B, D, E}; column δ allows {C, E}.

Worked solutions - Easy

Exercise 21 | Answer: E

Easy • 1 step in this route

	α	β	γ	δ	ε
1		B	A		
2	A	D	B		
3		A	C		E
4	C	**E**		A	B
5		C		B	

1. β4 = E: row 4 allows {D, E}; column β allows {E}.

Exercise 22 | Answer: D

Easy • 1 step in this route

	α	β	γ	δ	ε
1	B		**D**		E
2	D				A
3		C	E	D	
4	E	D	B	A	
5			A	B	

1. γ1 = D: in row 1, D fits only in column γ.

Exercise 23 | Answer: A

Easy • 1 step in this route

	α	β	γ	δ	ε
1		A	D		C
2	D	B		A	E
3	C		A		D
4	**A**				B
5				D	A

1. α4 = A: in row 4, A fits only in column α.

Exercise 24 | Answer: C

Easy • 1 step in this route

	α	β	γ	δ	ε
1	B	C			E
2	**C**	A	B		D
3	E	B	C		
4				C	B
5			A	B	

1. α2 = C: row 2 allows {C, E}; column α allows {A, C, D}.

Worked solutions - Easy

Exercise 25 | Answer: A

Easy • 1 step in this route

	α	β	γ	δ	ε
1		C	E		A
2	E	D		A	B
3	C		**A**		
4	A	E		B	
5		A	B		

1. γ3 = A: in row 3, A fits only in column γ.

Exercise 26 | Answer: D

Easy • 1 step in this route

	α	β	γ	δ	ε
1	B			E	A
2			E		D
3	A	B	C	**D**	
4		D	B		
5		E		C	

1. δ3 = D: row 3 allows {D, E}; column δ allows {A, B, D}.

Exercise 27 | Answer: A

Easy • 1 step in this route

	α	β	γ	δ	ε
1		C	B		
2	A		D		
3	B		C		
4			**A**		B
5	C	B	E	D	A

1. γ4 = A: row 4 allows {A, C, D, E}; column γ allows {A}.

Exercise 28 | Answer: B

Easy • 1 step in this route

	α	β	γ	δ	ε
1	A		D	E	
2	E				
3	B	D	E		
4	D		**B**	C	A
5				D	E

1. γ4 = B: row 4 allows {B, E}; column γ allows {A, B, C}.

Worked solutions - Easy

Exercise 29 | Answer: A

Easy • 1 step in this route

	α	β	γ	δ	ε
1				A	D
2	E		C		
3	B		A		C
4	D	E			**A**
5	A	C		B	

1. ε4 = A: in row 4, A fits only in column ε.

Exercise 30 | Answer: D

Easy • 1 step in this route

	α	β	γ	δ	ε
1		B			**D**
2	D	C			
3				D	
4	C	A	D	E	B
5		D		C	A

1. ε1 = D: in row 1, D fits only in column ε.

Exercise 31 | Answer: C

Easy • 1 step in this route

	α	β	γ	δ	ε
1			A		
2			**C**		
3	C	E	D		
4	B	C	E	A	
5	D		B	C	

1. γ2 = C: row 2 allows {A, B, C, D, E}; column γ allows {C}.

Exercise 32 | Answer: C

Easy • 1 step in this route

	α	β	γ	δ	ε
1					A
2	C		B	D	
3		E			B
4		C	A	B	
5	B			A	**C**

1. ε5 = C: in column ε, C fits only in row 5.

Worked solutions - Easy

Exercise 33 | Answer: D

Easy • 1 step in this route

	α	β	γ	δ	ε
1				A	B
2		B		C	**D**
3		D	C		
4	E	A			C
5	D	C			

1. ε2 = D: in column ε, D fits only in row 2.

Exercise 34 | Answer: A

Easy • 1 step in this route

	α	β	γ	δ	ε
1		A	E		
2	B				C
3	E		D	A	
4			C	E	A
5	**A**		B		

1. α5 = A: in row 5, A fits only in column α.

Exercise 35 | Answer: E

Easy • 1 step in this route

	α	β	γ	δ	ε
1		A			E
2			A		D
3		B	D	A	
4				E	
5	B	C	**E**	D	

1. γ5 = E: row 5 allows {A, E}; column γ allows {B, C, E}.

Exercise 36 | Answer: A

Easy • 2 steps in this route

	α	β	γ	δ	ε
1			A	E	B
2	E			D	
3		B	C	A	E
4		E		B	C
5	B	**A**	E	C	D

1. ε5 = D: in column ε, D fits only in row 5.

2. β5 = A: row 5 allows {A}; column β allows {A, C, D}.

Worked solutions - Easy

Exercise 37 | Answer: D

Easy • 2 steps in this route

	α	β	γ	δ	ε
1	C		A	D	E
2				C	B
3		E	D	A	C
4			C	B	D
5	**D**	C		E	A

1. β5 = C: in row 5, C fits only in column β.

2. α5 = D: in row 5, D fits only in column α.

Exercise 38 | Answer: D

Easy • 2 steps in this route

	α	β	γ	δ	ε
1	**D**	C		E	
2	E		C	D	
3	C	E	A	B	
4	A	B	D		E
5		D	E		C

1. β5 = D: in column β, D fits only in row 5.

2. α1 = D: in column α, D fits only in row 1.

Exercise 39 | Answer: D

Easy • 2 steps in this route

	α	β	γ	δ	ε
1	E	B			A
2	B		**D**		C
3	D	C	B	A	
4	A	D	E	C	B
5			A		D

1. ε2 = C: in column ε, C fits only in row 2.

2. γ2 = D: row 2 allows {A, D, E}; column γ allows {C, D}.

Exercise 40 | Answer: B

Easy • 2 steps in this route

	α	β	γ	δ	ε
1	D	E	C	B	A
2		B		C	
3	C				**B**
4	A	D		E	C
5	B		D	A	E

1. ε4 = C: in row 4, C fits only in column ε.

2. ε3 = B: in column ε, B fits only in row 3.

Worked solutions - Easy

Exercise 41 | Answer: E

Easy • 2 steps in this route

	α	β	γ	δ	ε
1				C	B
2	**E**		B	A	C
3	B	C		D	
4	A	B	C	E	
5	C	E			A

1. ε2 = C: in row 2, C fits only in column ε.

2. α2 = E: in row 2, E fits only in column α.

Exercise 42 | Answer: A

Easy • 2 steps in this route

	α	β	γ	δ	ε
1		D		B	
2		A	B	E	C
3	A	C			B
4	C	B		**A**	E
5	B	E		C	

1. δ5 = C: in column δ, C fits only in row 5.

2. δ4 = A: in column δ, A fits only in row 4.

Exercise 43 | Answer: E

Easy • 2 steps in this route

	α	β	γ	δ	ε
1		B	D		
2	A	**E**	C	B	D
3				A	C
4	D	A		C	
5	E		A	D	B

1. γ2 = C: in column γ, C fits only in row 2.

2. β2 = E: row 2 allows {E}; column β allows {C, D, E}.

Exercise 44 | Answer: D

Easy • 2 steps in this route

	α	β	γ	δ	ε
1	E		A	C	
2	A	B			C
3	C		B	A	**D**
4	B	A		D	E
5	D			B	

1. γ3 = B: in column γ, B fits only in row 3.

2. ε3 = D: row 3 allows {D, E}; column ε allows {A, B, D}.

Worked solutions - Easy

Exercise 45 | Answer: D

Easy • 2 steps in this route

	α	β	γ	δ	ε
1				C	**D**
2	C		D		B
3	E	D	C		A
4	A	C	B		E
5		B	A	E	

1. δ1 = C: in column δ, C fits only in row 1.

2. ε1 = D: row 1 allows {A, B, D, E}; column ε allows {C, D}.

Exercise 46 | Answer: B

Easy • 2 steps in this route

	α	β	γ	δ	ε
1	D	B	A	C	E
2		D	C		B
3	**B**		E		
4		E			D
5	E	C	D	B	

1. δ5 = B: in row 5, B fits only in column δ.

2. α3 = B: in row 3, B fits only in column α.

Exercise 47 | Answer: B

Easy • 2 steps in this route

	α	β	γ	δ	ε
1	B	D	E		C
2	C	B	D		A
3		C		D	**B**
4	A			C	
5	D		C		E

1. δ3 = D: in column δ, D fits only in row 3.

2. ε3 = B: row 3 allows {A, B, E}; column ε allows {B, D}.

Exercise 48 | Answer: D

Easy • 2 steps in this route

	α	β	γ	δ	ε
1	A	B	**D**		
2	D	C		A	
3	B	E	A		C
4	E		C	B	D
5			B		A

1. γ3 = A: in row 3, A fits only in column γ.

2. γ1 = D: in column γ, D fits only in row 1.

Worked solutions - Easy

Exercise 49 | Answer: B

Easy • 2 steps in this route

	α	β	γ	δ	ε
1	C	D		A	
2	A			E	B
3		**B**	A	C	
4	B	E	C	D	
5	D		E		C

1. γ3 = A: in column γ, A fits only in row 3.

2. β3 = B: row 3 allows {B, D, E}; column β allows {A, B, C}.

Exercise 50 | Answer: C

Easy • 2 steps in this route

	α	β	γ	δ	ε
1	A	B	D	E	
2	**C**		A	D	B
3	E	D			A
4	B	A			D
5				A	

1. δ2 = D: in column δ, D fits only in row 2.

2. α2 = C: row 2 allows {C, E}; column α allows {C, D}.

Exercise 51 | Answer: E

Easy • 2 steps in this route

	α	β	γ	δ	ε
1	D	A	B	E	
2					
3	**E**		C	A	D
4			A	B	E
5		E	D	C	B

1. β5 = E: row 5 allows {A, E}; column β allows {B, C, D, E}.

2. α3 = E: in row 3, E fits only in column α.

Exercise 52 | Answer: B

Easy • 2 steps in this route

	α	β	γ	δ	ε
1	E	C	B	D	A
2	D		E	C	
3	A	**B**	C		D
4		E		A	
5			A		

1. α3 = A: in column α, A fits only in row 3.

2. β3 = B: row 3 allows {B, E}; column β allows {A, B, D}.

Worked solutions - Easy

Exercise 53 | Answer: E

Easy • 2 steps in this route

	α	β	γ	δ	ε
1	C	E	D		A
2		C			**E**
3	E	D	B		C
4		A		E	
5	A			C	D

1. ε5 = D: in row 5, D fits only in column ε.

2. ε2 = E: in column ε, E fits only in row 2.

Exercise 54 | Answer: C

Easy • 2 steps in this route

	α	β	γ	δ	ε
1		E	A		C
2	C	B		E	
3		A	**C**	B	
4	D		B	A	E
5		D			B

1. β3 = A: in column β, A fits only in row 3.

2. γ3 = C: in row 3, C fits only in column γ.

Exercise 55 | Answer: C

Easy • 2 steps in this route

	α	β	γ	δ	ε
1	D	A	B		E
2	B	D			**C**
3	A		C	D	
4	E		D		A
5		B	A		

1. β2 = D: in column β, D fits only in row 2.

2. ε2 = C: row 2 allows {A, C, E}; column ε allows {B, C, D}.

Exercise 56 | Answer: A

Easy • 2 steps in this route

	α	β	γ	δ	ε
1		B	C	D	E
2		E	A	C	D
3		A	D		
4	D			B	A
5		D		**A**	

1. δ1 = D: in row 1, D fits only in column δ.

2. δ5 = A: in column δ, A fits only in row 5.

Worked solutions - Easy

Exercise 57 | Answer: C

Easy • 2 steps in this route

	α	β	γ	δ	ε
1		**C**		A	D
2	A		D	E	C
3	D				
4		E	C	D	A
5	C		A	B	

1. γ4 = C: in row 4, C fits only in column γ.

2. β1 = C: in row 1, C fits only in column β.

Exercise 58 | Answer: E

Easy • 2 steps in this route

	α	β	γ	δ	ε
1		D	C	A	
2	A			**E**	C
3		C	A		
4		A	E		D
5	C	E			A

1. ε2 = C: in column ε, C fits only in row 2.

2. δ2 = E: in row 2, E fits only in column δ.

Exercise 59 | Answer: B

Easy • 2 steps in this route

	α	β	γ	δ	ε
1	A		B		
2			D	E	
3		C		B	D
4		D	C	A	
5	D	A		C	**B**

1. α5 = D: in row 5, D fits only in column α.

2. ε5 = B: in row 5, B fits only in column ε.

Exercise 60 | Answer: A

Easy • 2 steps in this route

	α	β	γ	δ	ε
1	A	E	D		
2				D	C
3	C	D	B		A
4	E		A	B	
5	D			**A**	

1. δ2 = D: in row 2, D fits only in column δ.

2. δ5 = A: in column δ, A fits only in row 5.

Worked solutions - Easy

Exercise 61 | Answer: E

Easy • 2 steps in this route

	α	β	γ	δ	ε
1	D	A	C		
2		D	B	C	A
3	A	**E**		B	
4		C	A		E
5				A	

1. β2 = D: in row 2, D fits only in column β.

2. β3 = E: row 3 allows {C, D, E}; column β allows {B, E}.

Exercise 62 | Answer: D

Easy • 2 steps in this route

	α	β	γ	δ	ε
1			E	B	D
2	C	**D**		E	
3		E	D		C
4	D	B		C	E
5					B

1. δ2 = E: in row 2, E fits only in column δ.

2. β2 = D: in row 2, D fits only in column β.

Exercise 63 | Answer: C

Easy • 2 steps in this route

	α	β	γ	δ	ε
1		A			D
2		B	A		E
3	E				B
4		E	B	D	**C**
5		D	C	E	

1. ε1 = D: in column ε, D fits only in row 1.

2. ε4 = C: in column ε, C fits only in row 4.

Exercise 64 | Answer: C

Easy • 2 steps in this route

	α	β	γ	δ	ε
1			A	E	**C**
2			D	C	E
3		A			D
4		E		A	B
5	E		B	D	

1. γ1 = A: in column γ, A fits only in row 1.

2. ε1 = C: row 1 allows {B, C, D}; column ε allows {A, C}.

Worked solutions - Easy

Exercise 65 | Answer: B

Easy • 2 steps in this route

	α	β	γ	δ	ε
1	C	D	B		
2	E	**B**	A	C	
3			D	E	
4	D			B	
5		C	E		

1. α2 = E: in column α, E fits only in row 2.

2. β2 = B: row 2 allows {B, D}; column β allows {A, B, E}.

Exercise 66 | Answer: A

Easy • 2 steps in this route

	α	β	γ	δ	ε
1	**A**	E	B	D	
2		A		B	
3		B	E		
4	B				E
5	C	D			B

1. β1 = E: in column β, E fits only in row 1.

2. α1 = A: row 1 allows {A, C}; column α allows {A, D, E}.

Exercise 67 | Answer: C

Easy • 2 steps in this route

	α	β	γ	δ	ε
1	B	E	C	D	
2			E	**C**	B
3	A				C
4		D			E
5				A	D

1. ε2 = B: in column ε, B fits only in row 2.

2. δ2 = C: row 2 allows {A, C, D}; column δ allows {B, C, E}.

Exercise 68 | Answer: B

Easy • 2 steps in this route

	α	β	γ	δ	ε
1		C	**B**		A
2		A	C		
3	A		D		B
4	C	B			
5	E	D	A		

1. γ2 = C: in column γ, C fits only in row 2.

2. γ1 = B: in column γ, B fits only in row 1.

Worked solutions - Easy

Exercise 69 | Answer: D

Easy • 2 steps in this route

	α	β	γ	δ	ε
1	A	**D**	E		
2	D			E	
3		C			
4	B			D	C
5	C		D	B	E

1. α2 = D: row 2 allows {A, B, C, D}; column α allows {D, E}.

2. β1 = D: in column β, D fits only in row 1.

Exercise 70 | Answer: E

Easy • 2 steps in this route

	α	β	γ	δ	ε
1		B		**E**	C
2		E	C		
3	C				
4	E		B		A
5	A	D	E		B

1. ε1 = C: in column ε, C fits only in row 1.

2. δ1 = E: in row 1, E fits only in column δ.

Worked solutions - Moderate

Follow the numbered deductions. Magenta letters show the route to the shaded target. Other blank cells need not be filled.

Exercise 71 | Answer: D

Moderate • 3 steps in this route

	α	β	γ	δ	ε
1		B		C	
2			B	E	A
3	B	E	D	A	
4	E	A	C	**D**	
5	C	D	A		

1. **δ2 = E**: in row 2, E fits only in column δ.
2. **δ1 = C**: in column δ, C fits only in row 1.
3. **δ4 = D**: in column δ, D fits only in row 4.

Exercise 72 | Answer: E

Moderate • 3 steps in this route

	α	β	γ	δ	ε
1	B	D	C	E	A
2		C			B
3			B	C	
4	A	E	D	B	
5	C			D	**E**

1. **β2 = C**: in column β, C fits only in row 2.
2. **ε2 = B**: in row 2, B fits only in column ε.
3. **ε5 = E**: row 5 allows {A, B, E}; column ε allows {C, D, E}.

Exercise 73 | Answer: D

Moderate • 3 steps in this route

	α	β	γ	δ	ε
1		C	E	A	
2		**D**			A
3	B		D	C	E
4	A	E			C
5	C		A	E	D

1. **α4 = A**: in column α, A fits only in row 4.
2. **β4 = E**: in row 4, E fits only in column β.
3. **β2 = D**: in column β, D fits only in row 2.

Exercise 74 | Answer: A

Moderate • 3 steps in this route

	α	β	γ	δ	ε
1	E	A	B	D	C
2			D	E	
3	D		C	B	**A**
4	B		A		E
5	C				

1. **γ2 = D**: row 2 allows {A, B, C, D}; column γ allows {D, E}.
2. **α3 = D**: in column α, D fits only in row 3.
3. **ε3 = A**: row 3 allows {A, E}; column ε allows {A, B, D}.

Worked solutions - Moderate

Exercise 75 | Answer: E

Moderate • 3 steps in this route

	α	β	γ	δ	ε
1	B	E	A		
2				**E**	
3	E	D		C	A
4	D	C		A	B
5	C		D		E

1. β3 = D: in row 3, D fits only in column β.

2. β1 = E: row 1 allows {C, D, E}; column β allows {A, B, E}.

3. δ2 = E: in column δ, E fits only in row 2.

Exercise 76 | Answer: E

Moderate • 3 steps in this route

	α	β	γ	δ	ε
1	C	**E**	B		A
2	B	A		E	C
3			C		B
4	A			B	
5		B	A	C	

1. ε3 = B: in row 3, B fits only in column ε.

2. ε1 = A: in column ε, A fits only in row 1.

3. β1 = E: in row 1, E fits only in column β.

Exercise 77 | Answer: C

Moderate • 3 steps in this route

	α	β	γ	δ	ε
1		**C**		E	A
2	E				C
3	C	A	B		
4	D	E	C	A	B
5	A		E		

1. ε1 = A: in column ε, A fits only in row 1.

2. δ1 = E: in row 1, E fits only in column δ.

3. β1 = C: in row 1, C fits only in column β.

Exercise 78 | Answer: C

Moderate • 3 steps in this route

	α	β	γ	δ	ε
1			A		**C**
2	C		B	D	
3		B	E		D
4	E				B
5	D		C	B	E

1. γ3 = E: in column γ, E fits only in row 3.

2. ε3 = D: in row 3, D fits only in column ε.

3. ε1 = C: row 1 allows {B, C, D, E}; column ε allows {A, C}.

Worked solutions - Moderate

Exercise 79 | Answer: E

Moderate • 3 steps in this route

	α	β	γ	δ	ε
1				B	
2	D		B		E
3	**E**	D	A		
4			C	D	A
5	A	B		E	C

1. δ4 = D: in row 4, D fits only in column δ.

2. δ5 = E: row 5 allows {D, E}; column δ allows {A, C, E}.

3. α3 = E: in row 3, E fits only in column α.

Exercise 80 | Answer: D

Moderate • 3 steps in this route

	α	β	γ	δ	ε
1	B		E	D	A
2					
3	E	B	**D**		C
4		E		C	B
5	C			E	D

1. δ5 = E: in row 5, E fits only in column δ.

2. α3 = E: in row 3, E fits only in column α.

3. γ3 = D: in row 3, D fits only in column γ.

Exercise 81 | Answer: B

Moderate • 3 steps in this route

	α	β	γ	δ	ε
1		E	D	C	
2	E		**B**	A	
3	B	A	C		D
4	D			B	
5	C				E

1. δ1 = C: in column δ, C fits only in row 1.

2. γ1 = D: in row 1, D fits only in column γ.

3. γ2 = B: row 2 allows {B, C, D}; column γ allows {A, B, E}.

Exercise 82 | Answer: A

Moderate • 3 steps in this route

	α	β	γ	δ	ε
1	B		C		
2	E	C		A	
3		E	D	B	C
4		B	**A**	C	
5		A		D	

1. δ3 = B: in row 3, B fits only in column δ.

2. δ2 = A: row 2 allows {A, B, D}; column δ allows {A, E}.

3. γ4 = A: in column γ, A fits only in row 4.

Worked solutions - Moderate

Exercise 83 | Answer: C

Moderate • 3 steps in this route

	α	β	γ	δ	ε
1		B			C
2	B	**C**	E		
3	D			B	E
4		D		E	A
5		E		C	B

1. δ4 = E: row 4 allows {B, C, E}; column δ allows {A, D, E}.

2. γ2 = E: in row 2, E fits only in column γ.

3. β2 = C: in row 2, C fits only in column β.

Exercise 84 | Answer: D

Moderate • 3 steps in this route

	α	β	γ	δ	ε
1	A	B	E	D	
2		E			D
3	**D**	C	B		
4	E		D	C	
5			C		A

1. γ5 = C: row 5 allows {B, C, D, E}; column γ allows {A, C}.

2. β3 = C: in column β, C fits only in row 3.

3. α3 = D: row 3 allows {A, D, E}; column α allows {B, C, D}.

Exercise 85 | Answer: B

Moderate • 3 steps in this route

	α	β	γ	δ	ε
1		E	D	A	**B**
2			C		
3	B	C		D	A
4	D			C	
5		D	B		

1. δ3 = D: in row 3, D fits only in column δ.

2. α3 = B: in row 3, B fits only in column α.

3. ε1 = B: in row 1, B fits only in column ε.

Exercise 86 | Answer: E

Moderate • 3 steps in this route

	α	β	γ	δ	ε
1	B		A	E	
2	C		E		
3			C		D
4	**E**		D		A
5	D			A	

1. γ1 = A: in column γ, A fits only in row 1.

2. α1 = B: row 1 allows {B, C, D}; column α allows {A, B, E}.

3. α4 = E: row 4 allows {B, C, E}; column α allows {A, E}.

Worked solutions - Moderate

Exercise 87 | Answer: E

Moderate • 3 steps in this route

	α	β	γ	δ	ε
1	D			**E**	
2	E	B	D		
3			B		E
4	C			B	D
5	B			C	

1. **α5 = B**: in column α, B fits only in row 5.
2. **α2 = E**: in column α, E fits only in row 2.
3. **δ1 = E**: in column δ, E fits only in row 1.

Exercise 88 | Answer: B

Moderate • 4 steps in this route

	α	β	γ	δ	ε
1				**B**	C
2	D	B	E	C	A
3	A	C	B	D	
4	C	A		E	
5	B	E			D

1. **β5 = E**: row 5 allows {A, B, C, E}; column β allows {D, E}.
2. **α3 = A**: in row 3, A fits only in column α.
3. **α5 = B**: row 5 allows {A, B, C}; column α allows {B, E}.
4. **δ1 = B**: in row 1, B fits only in column δ.

Exercise 89 | Answer: E

Moderate • 4 steps in this route

	α	β	γ	δ	ε
1	E		C	B	D
2	B		A	E	C
3		**E**		A	
4	A				E
5	D	B		C	A

1. **δ1 = B**: in row 1, B fits only in column δ.
2. **δ3 = A**: in column δ, A fits only in row 3.
3. **δ2 = E**: in column δ, E fits only in row 2.
4. **β3 = E**: in column β, E fits only in row 3.

Exercise 90 | Answer: D

Moderate • 4 steps in this route

	α	β	γ	δ	ε
1	E	C		A	B
2		A	E	B	
3	D			C	
4	A	**D**		E	
5	B	E	C		

1. **β1 = C**: in row 1, C fits only in column β.
2. **δ3 = C**: in column δ, C fits only in row 3.
3. **α3 = D**: row 3 allows {A, B, D, E}; column α allows {C, D}.
4. **β4 = D**: in column β, D fits only in row 4.

Worked solutions - Moderate

Exercise 91 | Answer: A

Moderate • 4 steps in this route

	α	β	γ	δ	ε
1	E				
2	D	C	E		**A**
3	A		C	D	
4	B	E			D
5		A	D	E	B

1. **γ5 = D**: in row 5, D fits only in column γ.
2. **δ5 = E**: in column δ, E fits only in row 5.
3. **ε5 = B**: in row 5, B fits only in column ε.
4. **ε2 = A**: row 2 allows {A, B}; column ε allows {A, C, E}.

Exercise 92 | Answer: C

Moderate • 4 steps in this route

	α	β	γ	δ	ε
1	A	D		E	C
2				B	
3		B	E	C	
4	C	E		D	
5	B	**C**	D		E

1. **δ3 = C**: in row 3, C fits only in column δ.
2. **α5 = B**: in column α, B fits only in row 5.
3. **γ5 = D**: in row 5, D fits only in column γ.
4. **β5 = C**: in row 5, C fits only in column β.

Exercise 93 | Answer: C

Moderate • 4 steps in this route

	α	β	γ	δ	ε
1	E				
2	D	B	E	A	
3			B	D	E
4		E		**C**	B
5	B	D	C		A

1. **β5 = D**: row 5 allows {D, E}; column β allows {A, C, D}.
2. **ε3 = E**: in column ε, E fits only in row 3.
3. **δ3 = D**: in row 3, D fits only in column δ.
4. **δ4 = C**: row 4 allows {A, C, D}; column δ allows {B, C, E}.

Exercise 94 | Answer: A

Moderate • 4 steps in this route

	α	β	γ	δ	ε
1	**A**	D			B
2		E	B	C	
3	E	B	A	D	
4	C		D		E
5					D

1. **ε1 = B**: in column ε, B fits only in row 1.
2. **δ3 = D**: in column δ, D fits only in row 3.
3. **α3 = E**: row 3 allows {C, E}; column α allows {A, B, D, E}.
4. **α1 = A**: row 1 allows {A, C, E}; column α allows {A, B, D}.

Worked solutions - Moderate

Exercise 95 | Answer: A

Moderate • 4 steps in this route

	α	β	γ	δ	ε
1			E		
2	A		D		B
3		A	C		
4	D		B	A	E
5	E	B		D	

1. γ1 = E: in column γ, E fits only in row 1.
2. γ4 = B: in column γ, B fits only in row 4.
3. δ4 = A: in row 4, A fits only in column δ.
4. α2 = A: in row 2, A fits only in column α.

Exercise 96 | Answer: A

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	A	C	E	B	D
2	E	A	B		
3	B		C		A
4	C		D	A	E
5	D		A		

1. γ4 = D: row 4 allows {A, B, C, D}; column γ allows {D}.
2. α4 = C: row 4 allows {A, B, C}; column α allows {C}.
3. δ1 = B: row 1 allows {B}; column δ allows {A, B, C, D, E}.
4. δ4 = A: row 4 allows {A, B}; column δ allows {A, C, D, E}.
5. β2 = A: in row 2, A fits only in column β.

Exercise 97 | Answer: B

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	D	E	C		
2	B	D		C	
3	E	C	A	B	D
4	A	B	D	E	
5			B		E

1. γ4 = D: in column γ, D fits only in row 4.
2. δ4 = E: row 4 allows {C, E}; column δ allows {A, B, D, E}.
3. ε5 = E: in row 5, E fits only in column ε.
4. ε3 = D: in column ε, D fits only in row 3.
5. δ3 = B: row 3 allows {B}; column δ allows {A, B, D}.

Exercise 98 | Answer: E

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	D	B		C	
2		D	C		
3		C	A	E	D
4	C	E		A	B
5	E	A	B	D	

1. γ5 = B: in column γ, B fits only in row 5.
2. α5 = E: row 5 allows {C, D, E}; column α allows {A, B, E}.
3. δ5 = D: row 5 allows {C, D}; column δ allows {B, D, E}.
4. ε3 = D: in row 3, D fits only in column ε.
5. δ3 = E: in row 3, E fits only in column δ.

Worked solutions - Moderate

Exercise 99 | Answer: C

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	A	D			
2	D		B		A
3		B		A	E
4	E	A	**C**	B	
5	B	E		D	C

1. **α5 = B**: in row 5, B fits only in column α.
2. **α2 = D**: in row 2, D fits only in column α.
3. **α1 = A**: in column α, A fits only in row 1.
4. **α4 = E**: in column α, E fits only in row 4.
5. **γ4 = C**: in row 4, C fits only in column γ.

Exercise 100 | Answer: D

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	B	E	A		
2					C
3	D		C		A
4	C			A	E
5	A	C	E	**D**	B

1. **α4 = C**: in column α, C fits only in row 4.
2. **ε4 = E**: in row 4, E fits only in column ε.
3. **ε5 = B**: in column ε, B fits only in row 5.
4. **β5 = C**: in column β, C fits only in row 5.
5. **δ5 = D**: row 5 allows {D}; column δ allows {B, C, D, E}.

Exercise 101 | Answer: C

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	A	E	D	C	B
2	E		A		
3				E	
4	B		E	A	D
5	D		**C**	B	

1. **β1 = E**: row 1 allows {E}; column β allows {A, B, C, D, E}.
2. **δ4 = A**: row 4 allows {A, C, E}; column δ allows {A, B, D}.
3. **δ5 = B**: row 5 allows {A, B, C, E}; column δ allows {B, D}.
4. **γ4 = E**: in row 4, E fits only in column γ.
5. **γ5 = C**: row 5 allows {A, C, E}; column γ allows {B, C}.

Exercise 102 | Answer: E

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	A	B			D
2				B	**E**
3	E	C		D	
4		A	D		C
5	D		C	A	B

1. **δ3 = D**: row 3 allows {A, B, D}; column δ allows {C, D, E}.
2. **ε5 = B**: in row 5, B fits only in column ε.
3. **γ4 = D**: in row 4, D fits only in column γ.
4. **ε1 = D**: in row 1, D fits only in column ε.
5. **ε2 = E**: in column ε, E fits only in row 2.

Worked solutions - Moderate

Exercise 103 | Answer: C

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	C		E	B	D
2					E
3	E	**C**	B		A
4		E	D	C	B
5		B			

1. **δ1 = B**: in row 1, B fits only in column δ.
2. **β4 = E**: in column β, E fits only in row 4.
3. **δ4 = C**: in row 4, C fits only in column δ.
4. **γ3 = B**: in row 3, B fits only in column γ.
5. **β3 = C**: in row 3, C fits only in column β.

Exercise 104 | Answer: A

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	D		A	C	E
2	C		E		
3	E	C	B	**A**	
4		A	D		
5		E		D	

1. **α1 = D**: in column α, D fits only in row 1.
2. **β3 = C**: in column β, C fits only in row 3.
3. **γ1 = A**: in row 1, A fits only in column γ.
4. **γ3 = B**: row 3 allows {A, B, D}; column γ allows {B, C}.
5. **δ3 = A**: row 3 allows {A, D}; column δ allows {A, B, E}.

Exercise 105 | Answer: B

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	D	B	A		E
2	C	D	**B**		
3	E	A			
4	B	C	E	A	
5				D	

1. **γ4 = E**: in row 4, E fits only in column γ.
2. **β2 = D**: in column β, D fits only in row 2.
3. **α1 = D**: in column α, D fits only in row 1.
4. **γ1 = A**: in row 1, A fits only in column γ.
5. **γ2 = B**: row 2 allows {A, B, E}; column γ allows {B, C, D}.

Exercise 106 | Answer: D

Moderate • 4 steps in this route

	α	β	γ	δ	ε
1	B		A	C	D
2				E	A
3	A				B
4	**D**	A		B	E
5		B	D	A	

1. **ε2 = A**: in row 2, A fits only in column ε.
2. **ε3 = B**: in column ε, B fits only in row 3.
3. **δ4 = B**: in column δ, B fits only in row 4.
4. **α4 = D**: in row 4, D fits only in column α.

Worked solutions - Moderate

Exercise 107 | Answer: D

Moderate • 4 steps in this route

	α	β	γ	δ	ε
1	C		A	E	
2		C	D		E
3		A		C	D
4	E		C	**D**	A
5	D			A	

1. **α1 = C**: in column α, C fits only in row 1.
2. **γ1 = A**: in row 1, A fits only in column γ.
3. **γ2 = D**: in column γ, D fits only in row 2.
4. **δ4 = D**: in column δ, D fits only in row 4.

Exercise 108 | Answer: B

Moderate • 4 steps in this route

	α	β	γ	δ	ε
1		C	D	B	E
2	C				
3	**B**		E	A	C
4	E	B			
5		E	A	C	

1. **δ5 = C**: in row 5, C fits only in column δ.
2. **β5 = E**: in row 5, E fits only in column β.
3. **γ3 = E**: in row 3, E fits only in column γ.
4. **α3 = B**: in row 3, B fits only in column α.

Exercise 109 | Answer: A

Moderate • 4 steps in this route

	α	β	γ	δ	ε
1		C	E		D
2		A	D		E
3		B	A		
4			B		
5	E	D	C	**A**	B

1. **γ4 = B**: row 4 allows {A, B, C, D, E}; column γ allows {B}.
2. **β3 = B**: in column β, B fits only in row 3.
3. **ε5 = B**: in column ε, B fits only in row 5.
4. **δ5 = A**: row 5 allows {A}; column δ allows {A, B, C, D, E}.

Exercise 110 | Answer: A

Moderate • 4 steps in this route

	α	β	γ	δ	ε
1		A	B	D	C
2		B	A	C	
3	C		E		
4	B	C		E	
5	**A**		C		

1. **δ1 = D**: row 1 allows {D, E}; column δ allows {A, B, C, D}.
2. **δ2 = C**: row 2 allows {C, D, E}; column δ allows {A, B, C}.
3. **α3 = C**: in row 3, C fits only in column α.
4. **α5 = A**: in column α, A fits only in row 5.

Worked solutions - Moderate

Exercise 111 | Answer: E

Moderate • 4 steps in this route

	α	β	γ	δ	ε
1		C			A
2			D	A	**E**
3	D		A	C	B
4		A			D
5		D	B	E	

1. **γ3 = A**: in row 3, A fits only in column γ.
2. **δ3 = C**: in row 3, C fits only in column δ.
3. **δ5 = E**: row 5 allows {A, C, E}; column δ allows {B, D, E}.
4. **ε2 = E**: in column ε, E fits only in row 2.

Exercise 112 | Answer: C

Moderate • 4 steps in this route

	α	β	γ	δ	ε
1		B	D		E
2	E	C		D	
3	B		C		
4		A	B		C
5	**C**		E	B	

1. **β2 = C**: in row 2, C fits only in column β.
2. **β1 = B**: row 1 allows {A, B, C}; column β allows {B, D, E}.
3. **δ5 = B**: in column δ, B fits only in row 5.
4. **α5 = C**: in row 5, C fits only in column α.

Exercise 113 | Answer: C

Moderate • 4 steps in this route

	α	β	γ	δ	ε
1		A	C		B
2	B		E	C	
3			A	B	
4	A		D	E	**C**
5	D		B		

1. **γ2 = E**: row 2 allows {A, B, D, E}; column γ allows {E}.
2. **α2 = B**: row 2 allows {A, B, D}; column α allows {B, C, E}.
3. **ε1 = B**: in row 1, B fits only in column ε.
4. **ε4 = C**: row 4 allows {B, C}; column ε allows {A, C, D, E}.

Exercise 114 | Answer: C

Moderate • 4 steps in this route

	α	β	γ	δ	ε
1	D		B	C	
2					B
3	E	D			C
4	B		D	E	
5	**C**	B	E		D

1. **ε2 = B**: in column ε, B fits only in row 2.
2. **ε5 = D**: in column ε, D fits only in row 5.
3. **γ5 = E**: in row 5, E fits only in column γ.
4. **α5 = C**: in row 5, C fits only in column α.

Worked solutions - Moderate

Exercise 115 | Answer: D

Moderate • 4 steps in this route

	α	β	γ	δ	ε
1		E	A		C
2	A				
3	B	A			**D**
4	C	D	B		E
5				C	A

1. **β3 = A**: in column β, A fits only in row 3.
2. **γ1 = A**: in column γ, A fits only in row 1.
3. **ε1 = C**: in row 1, C fits only in column ε.
4. **ε3 = D**: row 3 allows {C, D, E}; column ε allows {B, D}.

Exercise 116 | Answer: C

Moderate • 4 steps in this route

	α	β	γ	δ	ε
1		B		E	
2			**C**	A	B
3	D	C	A		E
4	C	E			A
5	B				

1. **γ3 = A**: in row 3, A fits only in column γ.
2. **ε4 = A**: in row 4, A fits only in column ε.
3. **ε2 = B**: in column ε, B fits only in row 2.
4. **γ2 = C**: in row 2, C fits only in column γ.

Exercise 117 | Answer: D

Moderate • 4 steps in this route

	α	β	γ	δ	ε
1	C	D	B		
2	A		C		E
3		C		A	
4	E	A	**D**		
5	B	E			

1. **α1 = C**: in column α, C fits only in row 1.
2. **α2 = A**: in column α, A fits only in row 2.
3. **β4 = A**: in column β, A fits only in row 4.
4. **γ4 = D**: row 4 allows {B, C, D}; column γ allows {A, D, E}.

Exercise 118 | Answer: A

Moderate • 4 steps in this route

	α	β	γ	δ	ε
1					E
2	B	D		**A**	C
3	E	C	A		
4		B	C	E	D
5	C				

1. **β4 = B**: in row 4, B fits only in column β.
2. **β3 = C**: row 3 allows {B, C, D}; column β allows {A, C, E}.
3. **ε2 = C**: in column ε, C fits only in row 2.
4. **δ2 = A**: row 2 allows {A, E}; column δ allows {A, B, C, D}.

Worked solutions - Moderate

Exercise 119 | Answer: E

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1				E	A
2			A	C	D
3			**E**		C
4	A	C	D	B	
5	E	A	C	D	B

1. **α4 = A**: row 4 allows {A, E}; column α allows {A, B, C, D}.
2. **ε1 = A**: in column ε, A fits only in row 1.
3. **γ2 = A**: in row 2, A fits only in column γ.
4. **δ1 = E**: row 1 allows {B, C, D, E}; column δ allows {A, E}.
5. **γ3 = E**: in column γ, E fits only in row 3.

Exercise 120 | Answer: C

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	B		D		
2	D	B	C	E	
3		D			E
4			E	D	B
5	E	**C**	A	B	D

1. **α2 = D**: in row 2, D fits only in column α.
2. **β2 = B**: in row 2, B fits only in column β.
3. **α1 = B**: in row 1, B fits only in column α.
4. **α5 = E**: in column α, E fits only in row 5.
5. **β5 = C**: row 5 allows {C}; column β allows {A, C, E}.

Exercise 121 | Answer: D

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	E	D	A	B	
2	A	E	C		B
3				E	
4	D	C	B	A	
5			E	C	**D**

1. **ε2 = B**: in row 2, B fits only in column ε.
2. **δ5 = C**: in column δ, C fits only in row 5.
3. **γ4 = B**: in row 4, B fits only in column γ.
4. **γ5 = E**: in column γ, E fits only in row 5.
5. **ε5 = D**: in row 5, D fits only in column ε.

Exercise 122 | Answer: E

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	C	D	A		
2	B				C
3	E		B	C	D
4	A	B		D	
5		C	**E**	B	A

1. **α1 = C**: in row 1, C fits only in column α.
2. **α2 = B**: in column α, B fits only in row 2.
3. **β4 = B**: in column β, B fits only in row 4.
4. **β5 = C**: in column β, C fits only in row 5.
5. **γ5 = E**: in row 5, E fits only in column γ.

Worked solutions - Moderate

Exercise 123 | Answer: B

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	E			C	
2	**B**		C		
3	C	D	A	B	E
4	A	E	B		
5		C	E		B

1. **ε3 = E**: row 3 allows {E}; column ε allows {A, C, D, E}.
2. **β5 = C**: in row 5, C fits only in column β.
3. **β4 = E**: row 4 allows {C, D, E}; column β allows {A, B, E}.
4. **α1 = E**: in row 1, E fits only in column α.
5. **α2 = B**: in column α, B fits only in row 2.

Exercise 124 | Answer: B

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	D	A	B		
2	E	D		B	A
3	B		E	A	
4		E		D	**B**
5			D	E	

1. **γ5 = D**: in column γ, D fits only in row 5.
2. **α1 = D**: in column α, D fits only in row 1.
3. **β1 = A**: in row 1, A fits only in column β.
4. **β4 = E**: in column β, E fits only in row 4.
5. **ε4 = B**: in row 4, B fits only in column ε.

Exercise 125 | Answer: A

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	D	E	A		C
2		D	C		
3	C	B		A	
4	**A**		B		
5	E	A		C	B

1. **α3 = C**: row 3 allows {C, D, E}; column α allows {A, B, C}.
2. **γ1 = A**: in row 1, A fits only in column γ.
3. **γ2 = C**: in row 2, C fits only in column γ.
4. **γ4 = B**: in column γ, B fits only in row 4.
5. **α4 = A**: row 4 allows {A, C, D, E}; column α allows {A, B}.

Exercise 126 | Answer: E

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1		E	D		A
2	A	D	B	E	
3	E		A	C	D
4		C	E		
5		A			**E**

1. **β1 = E**: in row 1, E fits only in column β.
2. **β2 = D**: in row 2, D fits only in column β.
3. **β4 = C**: in column β, C fits only in row 4.
4. **γ4 = E**: row 4 allows {A, B, D, E}; column γ allows {C, E}.
5. **ε5 = E**: in row 5, E fits only in column ε.

Worked solutions - Moderate

Exercise 127 | Answer: B

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	A		C	D	
2	E		A	**B**	C
3					A
4	C			E	D
5	B	C	D		E

1. **γ5 = D**: in row 5, D fits only in column γ.
2. **ε5 = E**: in row 5, E fits only in column ε.
3. **ε3 = A**: in column ε, A fits only in row 3.
4. **ε2 = C**: in column ε, C fits only in row 2.
5. **δ2 = B**: row 2 allows {B, D}; column δ allows {A, B, C}.

Exercise 128 | Answer: B

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	C	A	**B**	E	
2		C	A	B	
3	E				C
4				C	B
5	B	E		D	A

1. **ε3 = C**: in column ε, C fits only in row 3.
2. **δ4 = C**: in column δ, C fits only in row 4.
3. **δ1 = E**: in column δ, E fits only in row 1.
4. **β1 = A**: in row 1, A fits only in column β.
5. **γ1 = B**: in row 1, B fits only in column γ.

Exercise 129 | Answer: A

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	D	C	A		B
2	A	D	B		
3	E	B			
4	C			B	**A**
5		A	E		C

1. **γ2 = B**: in column γ, B fits only in row 2.
2. **α2 = A**: row 2 allows {A, C, E}; column α allows {A, B, D}.
3. **α1 = D**: row 1 allows {A, D, E}; column α allows {B, D}.
4. **γ1 = A**: row 1 allows {A, E}; column γ allows {A, C, D}.
5. **ε4 = A**: in row 4, A fits only in column ε.

Exercise 130 | Answer: D

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1		A		B	D
2	A	C	D	E	
3	C	E	B		A
4	B	**D**	A		
5	D				

1. **α5 = D**: in column α, D fits only in row 5.
2. **α2 = A**: in column α, A fits only in row 2.
3. **δ2 = E**: row 2 allows {B, E}; column δ allows {A, C, D, E}.
4. **β3 = E**: in row 3, E fits only in column β.
5. **β4 = D**: row 4 allows {C, D, E}; column β allows {B, D}.

Worked solutions - Moderate

Exercise 131 | Answer: D

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1			C	E	
2		B		A	D
3		E	D	B	C
4	E	C		D	
5		**D**	A		E

1. **δ1 = E**: in row 1, E fits only in column δ.
2. **β4 = C**: in row 4, C fits only in column β.
3. **δ2 = A**: in column δ, A fits only in row 2.
4. **β2 = B**: row 2 allows {B, C, E}; column β allows {A, B, D}.
5. **β5 = D**: row 5 allows {B, C, D}; column β allows {A, D}.

Exercise 132 | Answer: E

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1			A	B	
2					B
3		D	B	E	C
4	B	**E**	D	C	A
5	C		E		D

1. **γ3 = B**: in column γ, B fits only in row 3.
2. **ε3 = C**: in row 3, C fits only in column ε.
3. **α4 = B**: in column α, B fits only in row 4.
4. **ε4 = A**: in column ε, A fits only in row 4.
5. **β4 = E**: row 4 allows {E}; column β allows {A, B, C, E}.

Exercise 133 | Answer: A

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	A	B	E	C	
2	B		D	E	
3	C	D	**A**		
4		E			
5	E		B	D	A

1. **δ5 = D**: in row 5, D fits only in column δ.
2. **β1 = B**: in column β, B fits only in row 1.
3. **δ1 = C**: row 1 allows {C, D, E}; column δ allows {A, B, C}.
4. **γ1 = E**: row 1 allows {D, E}; column γ allows {A, C, E}.
5. **γ3 = A**: row 3 allows {A, B, E}; column γ allows {A, C}.

Exercise 134 | Answer: B

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1		D		B	
2			D	E	
3	E		C		**B**
4		E	B	A	
5	D	B		C	E

1. **δ1 = B**: row 1 allows {A, B, C, E}; column δ allows {B, D}.
2. **β4 = E**: in column β, E fits only in row 4.
3. **γ4 = B**: row 4 allows {B, C, D}; column γ allows {A, B, E}.
4. **β5 = B**: in row 5, B fits only in column β.
5. **ε3 = B**: in row 3, B fits only in column ε.

Worked solutions - Moderate

Exercise 135 | Answer: D

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	E		**D**		
2	A			D	
3	C	D			E
4		A	C	E	
5	D	B	E		A

1. **γ5 = E**: in row 5, E fits only in column γ.
2. **α5 = D**: in row 5, D fits only in column α.
3. **α3 = C**: in column α, C fits only in row 3.
4. **β3 = D**: row 3 allows {A, B, D}; column β allows {C, D, E}.
5. **γ1 = D**: in column γ, D fits only in row 1.

Exercise 136 | Answer: B

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	B	E		C	
2		D	B	A	E
3		**B**		D	A
4	D	A	E		
5				E	

1. **β1 = E**: row 1 allows {A, D, E}; column β allows {B, C, E}.
2. **δ2 = A**: in column δ, A fits only in row 2.
3. **ε2 = E**: in column ε, E fits only in row 2.
4. **γ2 = B**: in row 2, B fits only in column γ.
5. **β3 = B**: in row 3, B fits only in column β.

Exercise 137 | Answer: B

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	**B**	C	E		A
2		A			
3				B	C
4	E	B	C	A	
5	C			E	B

1. **α5 = C**: in row 5, C fits only in column α.
2. **ε3 = C**: in row 3, C fits only in column ε.
3. **ε1 = A**: in column ε, A fits only in row 1.
4. **γ1 = E**: in row 1, E fits only in column γ.
5. **α1 = B**: in row 1, B fits only in column α.

Exercise 138 | Answer: C

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	D		E		
2	**C**	B		E	D
3			D		A
4	B				E
5	A	E	B		C

1. **γ1 = E**: in row 1, E fits only in column γ.
2. **γ5 = B**: in column γ, B fits only in row 5.
3. **ε5 = C**: row 5 allows {A, C, D}; column ε allows {B, C}.
4. **α5 = A**: row 5 allows {A, D}; column α allows {A, C, E}.
5. **α2 = C**: row 2 allows {A, C}; column α allows {C, E}.

Worked solutions - Moderate

Exercise 139 | Answer: A

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1		D		B	E
2	C	B	A		D
3			D		
4		A	B	D	
5	D			**A**	B

1. ε2 = D: in row 2, D fits only in column ε.

2. α5 = D: in row 5, D fits only in column α.

3. β2 = B: in row 2, B fits only in column β.

4. ε5 = B: in row 5, B fits only in column ε.

5. δ5 = A: in row 5, A fits only in column δ.

Exercise 140 | Answer: B

Moderate • 5 steps in this route

	α	β	γ	δ	ε
1	E	C		B	D
2	C	A		D	
3	B				A
4	D	B	C		
5			**B**		C

1. ε1 = D: row 1 allows {A, D}; column ε allows {B, C, D, E}.

2. ε5 = C: in column ε, C fits only in row 5.

3. α4 = D: in row 4, D fits only in column α.

4. α3 = B: row 3 allows {B, C, D, E}; column α allows {A, B}.

5. γ5 = B: in row 5, B fits only in column γ.

Worked solutions - High

Follow the numbered deductions. Magenta letters show the route to the shaded target. Other blank cells need not be filled.

Exercise 141 | Answer: D

High • 6 steps in this route

	α	β	γ	δ	ε
1	B	A	E		D
2	A	E			B
3	C	D	A	B	
4		C		E	A
5	E		**D**	A	C

1. **δ5 = A**: in row 5, A fits only in column δ.
2. **α5 = E**: in row 5, E fits only in column α.
3. **δ4 = E**: in row 4, E fits only in column δ.
4. **δ3 = B**: in column δ, B fits only in row 3.
5. **β3 = D**: row 3 allows {D, E}; column β allows {B, D}.
6. **γ5 = D**: in row 5, D fits only in column γ.

Exercise 142 | Answer: E

High • 6 steps in this route

	α	β	γ	δ	ε
1		C	B		
2	B	E	D		C
3	C		A	B	**E**
4	E	A	C	D	
5		B	E	C	D

1. **γ3 = A**: row 3 allows {A, B, D, E}; column γ allows {A}.
2. **α2 = B**: in column α, B fits only in row 2.
3. **β4 = A**: in column β, A fits only in row 4.
4. **δ4 = D**: in row 4, D fits only in column δ.
5. **δ3 = B**: in column δ, B fits only in row 3.
6. **ε3 = E**: row 3 allows {D, E}; column ε allows {A, B, E}.

Exercise 143 | Answer: B

High • 6 steps in this route

	α	β	γ	δ	ε
1	C	D	E	A	B
2					E
3	E	C	A	**B**	
4		A	B	E	C
5		E		C	A

1. **δ1 = A**: row 1 allows {A, C}; column δ allows {A, B, D}.
2. **α1 = C**: row 1 allows {C}; column α allows {A, B, C, D}.
3. **ε4 = C**: in row 4, C fits only in column ε.
4. **γ3 = A**: in row 3, A fits only in column γ.
5. **β3 = C**: in row 3, C fits only in column β.
6. **δ3 = B**: in row 3, B fits only in column δ.

Exercise 144 | Answer: B

High • 6 steps in this route

	α	β	γ	δ	ε
1	C	A		**B**	D
2		E	D		
3		B	C	D	A
4	A		B	C	E
5		C			B

1. **ε3 = A**: in row 3, A fits only in column ε.
2. **ε4 = E**: in column ε, E fits only in row 4.
3. **γ4 = B**: row 4 allows {B, C, D}; column γ allows {A, B, E}.
4. **δ4 = C**: row 4 allows {C, D}; column δ allows {A, B, C, E}.
5. **α1 = C**: in row 1, C fits only in column α.
6. **δ1 = B**: in row 1, B fits only in column δ.

Worked solutions - High

Exercise 145 | Answer: A

High • 6 steps in this route

	α	β	γ	δ	ε
1	B	D	C	A	
2	C				D
3		B	**A**	D	
4	A			E	B
5	D	E		C	A

1. **β1 = D**: in row 1, D fits only in column β.
2. **δ5 = C**: in column δ, C fits only in row 5.
3. **β5 = E**: row 5 allows {B, D, E}; column β allows {A, C, E}.
4. **α5 = D**: row 5 allows {B, D}; column α allows {A, D, E}.
5. **α4 = A**: row 4 allows {A, C, D}; column α allows {A, E}.
6. **γ3 = A**: in row 3, A fits only in column γ.

Exercise 146 | Answer: D

High • 6 steps in this route

	α	β	γ	δ	ε
1	B	C		E	**D**
2	E		B	D	
3		E	C	B	
4			D		E
5	C	D		A	B

1. **δ1 = E**: in column δ, E fits only in row 1.
2. **δ2 = D**: in column δ, D fits only in row 2.
3. **β5 = D**: in column β, D fits only in row 5.
4. **ε5 = B**: row 5 allows {B, E}; column ε allows {A, B, C, D}.
5. **α1 = B**: in row 1, B fits only in column α.
6. **ε1 = D**: in row 1, D fits only in column ε.

Exercise 147 | Answer: C

High • 6 steps in this route

	α	β	γ	δ	ε
1	A	E	C	D	B
2		A	B	C	D
3	D	B	E		C
4					
5	B	**C**		E	

1. **γ1 = C**: row 1 allows {C}; column γ allows {A, B, C, D}.
2. **γ2 = B**: row 2 allows {B, C, E}; column γ allows {A, B, D}.
3. **δ2 = C**: row 2 allows {C, E}; column δ allows {A, B, C}.
4. **α5 = B**: in row 5, B fits only in column α.
5. **ε3 = C**: in row 3, C fits only in column ε.
6. **β5 = C**: in row 5, C fits only in column β.

Exercise 148 | Answer: B

High • 6 steps in this route

	α	β	γ	δ	ε
1	A	D		C	
2	C	B			A
3		E	C	A	D
4	E	C			**B**
5	D	A			C

1. **β3 = E**: row 3 allows {A, B, C, D, E}; column β allows {E}.
2. **γ3 = C**: in row 3, C fits only in column γ.
3. **δ3 = A**: in row 3, A fits only in column δ.
4. **α4 = E**: in column α, E fits only in row 4.
5. **ε3 = D**: in row 3, D fits only in column ε.
6. **ε4 = B**: row 4 allows {A, B, D}; column ε allows {B, E}.

Worked solutions - High

Exercise 149 | Answer: E

High • 6 steps in this route

	α	β	γ	δ	ε
1			E	D	C
2		E		B	A
3	B	C	A	E	
4	**E**				B
5	C	D	B		E

1. **γ5 = B**: row 5 allows {A, B}; column γ allows {B, C, D, E}.
2. **ε4 = B**: in column ε, B fits only in row 4.
3. **γ1 = E**: row 1 allows {A, B, E}; column γ allows {C, D, E}.
4. **δ2 = B**: in column δ, B fits only in row 2.
5. **β2 = E**: row 2 allows {C, D, E}; column β allows {A, B, E}.
6. **α4 = E**: in row 4, E fits only in column α.

Exercise 150 | Answer: C

High • 6 steps in this route

	α	β	γ	δ	ε
1	B		D		C
2	C	D	A		E
3	D	C	E	A	B
4			C		A
5				**C**	

1. **ε2 = E**: row 2 allows {A, B, E}; column ε allows {C, D, E}.
2. **γ3 = E**: row 3 allows {E}; column γ allows {A, B, D, E}.
3. **γ2 = A**: in row 2, A fits only in column γ.
4. **γ1 = D**: row 1 allows {A, C, D, E}; column γ allows {B, D}.
5. **ε1 = C**: row 1 allows {A, C, E}; column ε allows {C, D}.
6. **δ5 = C**: in row 5, C fits only in column δ.

Exercise 151 | Answer: D

High • 6 steps in this route

	α	β	γ	δ	ε
1	C	**D**			B
2	A		D		C
3		C	B		
4	D		C	B	E
5	B	E	A		

1. **γ3 = B**: in column γ, B fits only in row 3.
2. **γ4 = C**: in column γ, C fits only in row 4.
3. **ε4 = E**: in row 4, E fits only in column ε.
4. **ε2 = C**: row 2 allows {B, C, E}; column ε allows {A, C, D}.
5. **β3 = C**: in column β, C fits only in row 3.
6. **β1 = D**: in column β, D fits only in row 1.

Exercise 152 | Answer: B

High • 6 steps in this route

	α	β	γ	δ	ε
1		E	B		A
2		C			E
3	E		D	A	C
4	B	A	C		D
5			E		**B**

1. **γ4 = C**: in column γ, C fits only in row 4.
2. **ε4 = D**: row 4 allows {A, D, E}; column ε allows {B, C, D}.
3. **β4 = A**: row 4 allows {A, E}; column β allows {A, B, D}.
4. **δ3 = A**: in row 3, A fits only in column δ.
5. **ε3 = C**: in row 3, C fits only in column ε.
6. **ε5 = B**: row 5 allows {A, B, C, D}; column ε allows {B}.

Worked solutions - High

Exercise 153 | Answer: A

High • 6 steps in this route

	α	β	γ	δ	ε
1		C	B	D	
2	C		A	B	E
3		B	E		
4		**A**		E	
5	A	E	D	C	

1. **ε2 = E**: in row 2, E fits only in column ε.
2. **δ2 = B**: in row 2, B fits only in column δ.
3. **δ5 = C**: row 5 allows {B, C, D}; column δ allows {A, C}.
4. **γ5 = D**: row 5 allows {B, D}; column γ allows {A, C, D}.
5. **γ2 = A**: row 2 allows {A, D}; column γ allows {A, C}.
6. **β4 = A**: in column β, A fits only in row 4.

Exercise 154 | Answer: C

High • 6 steps in this route

	α	β	γ	δ	ε
1		A	C	B	D
2	B	D		E	C
3	D		E		
4		**C**	B		E
5					B

1. **ε1 = D**: row 1 allows {B, C, D, E}; column ε allows {A, D}.
2. **γ1 = C**: row 1 allows {B, C, E}; column γ allows {A, C, D}.
3. **δ1 = B**: row 1 allows {B, E}; column δ allows {A, B, C, D}.
4. **α2 = B**: in column α, B fits only in row 2.
5. **β2 = D**: row 2 allows {A, D}; column β allows {B, C, D, E}.
6. **β4 = C**: row 4 allows {A, C, D}; column β allows {B, C, E}.

Exercise 155 | Answer: D

High • 6 steps in this route

	α	β	γ	δ	ε
1	E			B	D
2	C		B		A
3	D	B		A	C
4	A		**D**		
5		A	C		

1. **α3 = D**: row 3 allows {D, E}; column α allows {A, B, C, D}.
2. **ε1 = D**: row 1 allows {A, C, D}; column ε allows {B, D, E}.
3. **α4 = A**: in column α, A fits only in row 4.
4. **α2 = C**: in column α, C fits only in row 2.
5. **γ2 = B**: in row 2, B fits only in column γ.
6. **γ4 = D**: in column γ, D fits only in row 4.

Exercise 156 | Answer: C

High • 7 steps in this route

	α	β	γ	δ	ε
1	C	B	D	A	E
2		E	C	B	A
3	B	D			C
4	E		B	**C**	
5	A	C			

1. **β2 = E**: row 2 allows {D, E}; column β allows {A, B, C, E}.
2. **α5 = A**: in column α, A fits only in row 5.
3. **β1 = B**: row 1 allows {B, E}; column β allows {A, B, C}.
4. **ε1 = E**: row 1 allows {E}; column ε allows {B, C, D, E}.
5. **β5 = C**: row 5 allows {B, C, D, E}; column β allows {A, C}.
6. **ε3 = C**: row 3 allows {A, C, E}; column ε allows {B, C, D}.
7. **δ4 = C**: in row 4, C fits only in column δ.

Worked solutions - High

Exercise 157 | Answer: E

High • 7 steps in this route

	α	β	γ	δ	ε
1		D	C		A
2	C	B		A	D
3	D	C	A		**E**
4	E		B	D	C
5		E		C	

1. **δ4 = D**: row 4 allows {A, D}; column δ allows {B, D, E}.
2. **α2 = C**: in row 2, C fits only in column α.
3. **β2 = B**: in row 2, B fits only in column β.
4. **α3 = D**: in row 3, D fits only in column α.
5. **β1 = D**: in row 1, D fits only in column β.
6. **β5 = E**: in column β, E fits only in row 5.
7. **ε3 = E**: in column ε, E fits only in row 3.

Exercise 158 | Answer: C

High • 7 steps in this route

	α	β	γ	δ	ε
1		E		D	A
2	E	D			C
3	**C**	A		B	D
4	A		D	C	
5		C	A	E	B

1. **ε5 = B**: row 5 allows {A, B, C, D}; column ε allows {B, E}.
2. **β5 = C**: row 5 allows {A, C, D}; column β allows {B, C}.
3. **γ5 = A**: row 5 allows {A, D}; column γ allows {A, B, C, E}.
4. **α4 = A**: in column α, A fits only in row 4.
5. **δ4 = C**: in row 4, C fits only in column δ.
6. **δ3 = B**: row 3 allows {B, C, E}; column δ allows {A, B}.
7. **α3 = C**: row 3 allows {C, E}; column α allows {B, C, D}.

Exercise 159 | Answer: E

High • 7 steps in this route

	α	β	γ	δ	ε
1	A	**E**	B		C
2	B	C	A	E	D
3	D	A		B	E
4	E	B	D		
5	C				

1. **α1 = A**: row 1 allows {A, C, D, E}; column α allows {A}.
2. **δ3 = B**: in row 3, B fits only in column δ.
3. **β3 = A**: in row 3, A fits only in column β.
4. **ε1 = C**: row 1 allows {C, D, E}; column ε allows {A, B, C}.
5. **β2 = C**: in column β, C fits only in row 2.
6. **δ2 = E**: row 2 allows {E}; column δ allows {A, C, D, E}.
7. **β1 = E**: in row 1, E fits only in column β.

Exercise 160 | Answer: A

High • 7 steps in this route

	α	β	γ	δ	ε
1	C	E			B
2	B	**A**		E	D
3	D	B		A	C
4		D	B	C	
5	A				E

1. **α5 = A**: row 5 allows {A, B, C, D}; column α allows {A, E}.
2. **δ4 = C**: in row 4, C fits only in column δ.
3. **ε1 = B**: in column ε, B fits only in row 1.
4. **ε2 = D**: in column ε, D fits only in row 2.
5. **δ2 = E**: row 2 allows {A, C, E}; column δ allows {B, D, E}.
6. **β1 = E**: in column β, E fits only in row 1.
7. **β2 = A**: in column β, A fits only in row 2.

Worked solutions - High

Exercise 161 | Answer: C

High • 7 steps in this route

	α	β	γ	δ	ε
1	E	B	**C**	A	D
2	D			B	E
3		A	E	D	
4	A				B
5		D		E	A

1. **δ5 = E**: in row 5, E fits only in column δ.
2. **δ3 = D**: in row 3, D fits only in column δ.
3. **ε5 = A**: in column ε, A fits only in row 5.
4. **δ2 = B**: in column δ, B fits only in row 2.
5. **α4 = A**: in column α, A fits only in row 4.
6. **δ1 = A**: in column δ, A fits only in row 1.
7. **γ1 = C**: row 1 allows {C}; column γ allows {A, B, C, D}.

Exercise 162 | Answer: A

High • 7 steps in this route

	α	β	γ	δ	ε
1	B		C	E	D
2	D	B	A	C	E
3		E			
4		D		A	
5	E	C	D		**A**

1. **δ2 = C**: row 2 allows {A, C}; column δ allows {B, C, D, E}.
2. **γ2 = A**: row 2 allows {A}; column γ allows {A, B, C, E}.
3. **δ1 = E**: row 1 allows {A, C, E}; column δ allows {B, D, E}.
4. **γ1 = C**: row 1 allows {A, C}; column γ allows {B, C, E}.
5. **β5 = C**: in column β, C fits only in row 5.
6. **α5 = E**: in row 5, E fits only in column α.
7. **ε5 = A**: in row 5, A fits only in column ε.

Exercise 163 | Answer: C

High • 7 steps in this route

	α	β	γ	δ	ε
1	A	**C**			B
2		E		D	A
3	D	A	C		E
4	E		A		
5	B	D	E		C

1. **α3 = D**: in column α, D fits only in row 3.
2. **ε3 = E**: row 3 allows {B, E}; column ε allows {C, D, E}.
3. **ε5 = C**: row 5 allows {A, B, C, E}; column ε allows {C, D}.
4. **α5 = B**: row 5 allows {A, B, E}; column α allows {B, C}.
5. **γ5 = E**: row 5 allows {A, E}; column γ allows {B, D, E}.
6. **β2 = E**: in row 2, E fits only in column β.
7. **β1 = C**: row 1 allows {C, D, E}; column β allows {B, C}.

Exercise 164 | Answer: D

High • 7 steps in this route

	α	β	γ	δ	ε
1			**D**	C	E
2	C	E			A
3	E	C	A		
4	B	D	E		C
5	D	A		E	

1. **α2 = C**: row 2 allows {B, C, D}; column α allows {A, C}.
2. **δ5 = E**: in row 5, E fits only in column δ.
3. **β5 = A**: in row 5, A fits only in column β.
4. **β4 = D**: row 4 allows {A, C, D}; column β allows {B, D}.
5. **ε4 = C**: row 4 allows {A, C}; column ε allows {B, C, D}.
6. **δ1 = C**: in column δ, C fits only in row 1.
7. **γ1 = D**: in row 1, D fits only in column γ.

Worked solutions - High

Exercise 165 | Answer: C

High • 7 steps in this route

	α	β	γ	δ	ε
1	E	A	B		C
2	D		**C**	B	A
3			E	A	D
4	B	D		C	E
5	A				

1. **δ3 = A**: row 3 allows {A, B, C}; column δ allows {A, D, E}.
2. **ε4 = E**: in row 4, E fits only in column ε.
3. **ε2 = A**: row 2 allows {A, C, D, E}; column ε allows {A, B}.
4. **α4 = B**: in row 4, B fits only in column α.
5. **α5 = A**: in column α, A fits only in row 5.
6. **α2 = D**: in column α, D fits only in row 2.
7. **γ2 = C**: row 2 allows {C, E}; column γ allows {A, C, D}.

Exercise 166 | Answer: B

High • 7 steps in this route

	α	β	γ	δ	ε
1				A	D
2	D			**B**	
3	C		A	E	B
4	A	E	B		C
5	B	A	D		E

1. **γ4 = B**: row 4 allows {B, D}; column γ allows {A, B, C, E}.
2. **γ3 = A**: row 3 allows {A, B, D}; column γ allows {A, C, E}.
3. **ε3 = B**: row 3 allows {B, D}; column ε allows {A, B, E}.
4. **ε5 = E**: row 5 allows {B, C, E}; column ε allows {A, E}.
5. **α5 = B**: row 5 allows {B, C}; column α allows {B, E}.
6. **δ1 = A**: in row 1, A fits only in column δ.
7. **δ2 = B**: in column δ, B fits only in row 2.

Exercise 167 | Answer: E

High • 7 steps in this route

	α	β	γ	δ	ε
1	B		D	C	E
2			E		B
3		B		D	C
4		**E**	C		D
5		D	B	E	

1. **γ5 = B**: in row 5, B fits only in column γ.
2. **β3 = B**: in row 3, B fits only in column β.
3. **γ4 = C**: in column γ, C fits only in row 4.
4. **γ1 = D**: in column γ, D fits only in row 1.
5. **ε4 = D**: in column ε, D fits only in row 4.
6. **ε1 = E**: in column ε, E fits only in row 1.
7. **β4 = E**: in column β, E fits only in row 4.

Exercise 168 | Answer: B

High • 8 steps in this route

	α	β	γ	δ	ε
1	A			B	
2	B	E			C
3		D	**B**	E	A
4	D	B	C	A	E
5	E	A		C	

1. **β5 = A**: in column β, A fits only in row 5.
2. **α5 = E**: in column α, E fits only in row 5.
3. **β4 = B**: in column β, B fits only in row 4.
4. **α4 = D**: in column α, D fits only in row 4.
5. **γ4 = C**: row 4 allows {C}; column γ allows {A, B, C, D, E}.
6. **ε2 = C**: in row 2, C fits only in column ε.
7. **ε3 = A**: in column ε, A fits only in row 3.
8. **γ3 = B**: row 3 allows {B, C}; column γ allows {A, B, D, E}.

Worked solutions - High

Exercise 169 | Answer: B

High • 8 steps in this route

	α	β	γ	δ	ε
1	C	A	B		D
2		E	A		**B**
3	E		D	A	C
4	A	D			E
5		C	E		

1. **α3 = E**: in row 3, E fits only in column α.
2. **β5 = C**: in column β, C fits only in row 5.
3. **β1 = A**: in column β, A fits only in row 1.
4. **γ3 = D**: in row 3, D fits only in column γ.
5. **γ1 = B**: row 1 allows {B, D, E}; column γ allows {A, B, C}.
6. **ε1 = D**: row 1 allows {D, E}; column ε allows {A, B, D}.
7. **γ2 = A**: in column γ, A fits only in row 2.
8. **ε2 = B**: row 2 allows {B, C, D}; column ε allows {A, B}.

Exercise 170 | Answer: D

High • 9 steps in this route

	α	β	γ	δ	ε
1	C		A	E	
2	**D**		B	C	E
3	E	B		D	A
4			D	B	C
5		C	E	A	D

1. **δ2 = C**: row 2 allows {A, C, D, E}; column δ allows {C}.
2. **ε3 = A**: in row 3, A fits only in column ε.
3. **γ4 = D**: in column γ, D fits only in row 4.
4. **γ5 = E**: in column γ, E fits only in row 5.
5. **ε2 = E**: row 2 allows {A, D, E}; column ε allows {B, C, E}.
6. **β5 = C**: row 5 allows {B, C}; column β allows {A, C, D, E}.
7. **ε4 = C**: row 4 allows {A, C, E}; column ε allows {B, C}.
8. **α1 = C**: in row 1, C fits only in column α.
9. **α2 = D**: in column α, D fits only in row 2.

Exercise 171 | Answer: B

High • 6 steps in this route

	α	β	γ	δ	ε
1	**B**	A			D
2			C	A	E
3	E	C		B	A
4		E	A	D	
5	A		B	E	C

1. **γ4 = A**: in row 4, A fits only in column γ.
2. **β4 = E**: in row 4, E fits only in column β.
3. **δ5 = E**: in row 5, E fits only in column δ.
4. **ε5 = C**: in row 5, C fits only in column ε.
5. **ε1 = D**: in column ε, D fits only in row 1.
6. **α1 = B**: in row 1, B fits only in column α.

Exercise 172 | Answer: D

High • 6 steps in this route

	α	β	γ	δ	ε
1	B		C		A
2			D	C	E
3	E	C		A	D
4	**D**		A	B	C
5		A	E		

1. **ε1 = A**: in row 1, A fits only in column ε.
2. **ε2 = E**: in column ε, E fits only in row 2.
3. **δ2 = C**: row 2 allows {A, B, C}; column δ allows {C, D, E}.
4. **β3 = C**: in column β, C fits only in row 3.
5. **α3 = E**: row 3 allows {B, E}; column α allows {A, C, D, E}.
6. **α4 = D**: row 4 allows {D, E}; column α allows {A, C, D}.

Worked solutions - High

Exercise 173 | Answer: E

High • 6 steps in this route

	α	β	γ	δ	ε
1		C	D	A	E
2	C	A	E	B	D
3		D			
4			B	C	A
5	**E**	B		D	

1. **α2 = C**: row 2 allows {C}; column α allows {A, B, C, D, E}.
2. **β1 = C**: in row 1, C fits only in column β.
3. **δ4 = C**: in row 4, C fits only in column δ.
4. **β5 = B**: in column β, B fits only in row 5.
5. **δ5 = D**: in column δ, D fits only in row 5.
6. **α5 = E**: in row 5, E fits only in column α.

Exercise 174 | Answer: C

High • 6 steps in this route

	α	β	γ	δ	ε
1	E		A	D	B
2	B				
3	D	B	E		
4	A		B	C	D
5	**C**	A		B	E

1. **δ5 = B**: in row 5, B fits only in column δ.
2. **β3 = B**: in row 3, B fits only in column β.
3. **ε5 = E**: in row 5, E fits only in column ε.
4. **ε4 = D**: row 4 allows {D, E}; column ε allows {A, C, D}.
5. **α3 = D**: in row 3, D fits only in column α.
6. **α5 = C**: row 5 allows {C, D}; column α allows {C}.

Exercise 175 | Answer: E

High • 6 steps in this route

	α	β	γ	δ	ε
1			E	C	B
2		E	A	D	C
3	**E**	C			D
4	D	B	C		
5		A	D		E

1. **γ5 = D**: in row 5, D fits only in column γ.
2. **γ4 = C**: in column γ, C fits only in row 4.
3. **ε2 = C**: in column ε, C fits only in row 2.
4. **ε1 = B**: in column ε, B fits only in row 1.
5. **γ1 = E**: row 1 allows {A, D, E}; column γ allows {B, E}.
6. **α3 = E**: in column α, E fits only in row 3.

Exercise 176 | Answer: C

High • 6 steps in this route

	α	β	γ	δ	ε
1	B	C	A	E	
2	**C**		B	D	A
3			D		C
4	D	A			E
5		D	E		B

1. **ε4 = E**: in row 4, E fits only in column ε.
2. **β1 = C**: in row 1, C fits only in column β.
3. **ε5 = B**: in column ε, B fits only in row 5.
4. **β5 = D**: row 5 allows {A, C, D}; column β allows {B, D, E}.
5. **δ2 = D**: in row 2, D fits only in column δ.
6. **α2 = C**: in row 2, C fits only in column α.

Worked solutions - High

Exercise 177 | Answer: B

High • 6 steps in this route

	α	β	γ	δ	ε
1			D	C	A
2	E	B			
3	D	C	**B**		E
4	A	D		E	
5		A		B	D

1. **β5 = A**: in column β, A fits only in row 5.
2. **β4 = D**: in column β, D fits only in row 4.
3. **ε5 = D**: in row 5, D fits only in column ε.
4. **δ4 = E**: row 4 allows {B, C, E}; column δ allows {A, D, E}.
5. **ε3 = E**: in column ε, E fits only in row 3.
6. **γ3 = B**: in row 3, B fits only in column γ.

Exercise 178 | Answer: B

High • 6 steps in this route

	α	β	γ	δ	ε
1	C	E		**B**	D
2	E	A	B		C
3			E	C	A
4		D			
5	A	C	D		

1. **β2 = A**: in column β, A fits only in row 2.
2. **β5 = C**: in column β, C fits only in row 5.
3. **ε2 = C**: in row 2, C fits only in column ε.
4. **γ5 = D**: row 5 allows {B, D, E}; column γ allows {A, C, D}.
5. **ε1 = D**: in column ε, D fits only in row 1.
6. **δ1 = B**: in row 1, B fits only in column δ.

Exercise 179 | Answer: B

High • 6 steps in this route

	α	β	γ	δ	ε
1			B	D	
2	A	E	D	B	
3	**B**		A		D
4		D	C	A	B
5	D			C	

1. **γ2 = D**: in row 2, D fits only in column γ.
2. **δ4 = A**: in column δ, A fits only in row 4.
3. **γ4 = C**: in column γ, C fits only in row 4.
4. **β4 = D**: row 4 allows {D, E}; column β allows {A, B, C, D}.
5. **α5 = D**: in row 5, D fits only in column α.
6. **α3 = B**: in column α, B fits only in row 3.

Exercise 180 | Answer: D

High • 6 steps in this route

	α	β	γ	δ	ε
1		E	D		A
2	D	C			E
3	E	**D**		C	B
4	B				
5	A	B	C	E	

1. **β1 = E**: in row 1, E fits only in column β.
2. **α5 = A**: in column α, A fits only in row 5.
3. **β5 = B**: in column β, B fits only in row 5.
4. **γ5 = C**: row 5 allows {C, D}; column γ allows {A, B, C, E}.
5. **δ3 = C**: in row 3, C fits only in column δ.
6. **β3 = D**: in row 3, D fits only in column β.

Worked solutions - High

Exercise 181 | Answer: A

High • 6 steps in this route

	α	β	γ	δ	ε
1	D		E	C	
2		E	B		C
3	E			B	D
4	B	C		A	
5	C	D	**A**		

1. **α3 = E**: in row 3, E fits only in column α.
2. **α4 = B**: row 4 allows {B, C, D, E}; column α allows {A, B}.
3. **β4 = C**: row 4 allows {C, D, E}; column β allows {A, B, C}.
4. **ε2 = C**: in column ε, C fits only in row 2.
5. **γ2 = B**: in row 2, B fits only in column γ.
6. **γ5 = A**: row 5 allows {A, B, E}; column γ allows {A, C, D}.

Exercise 182 | Answer: D

High • 6 steps in this route

	α	β	γ	δ	ε
1	B			A	**D**
2		B	A		C
3	A				B
4	D	A	B	C	
5	C	E			A

1. **δ1 = A**: in row 1, A fits only in column δ.
2. **γ4 = B**: in row 4, B fits only in column γ.
3. **α1 = B**: in row 1, B fits only in column α.
4. **α5 = C**: row 5 allows {B, C, D}; column α allows {C, E}.
5. **ε2 = C**: in row 2, C fits only in column ε.
6. **ε1 = D**: in column ε, D fits only in row 1.

Exercise 183 | Answer: E

High • 6 steps in this route

	α	β	γ	δ	ε
1	E		D	A	
2	D	**E**	C	B	A
3			B		D
4	C	A	E		B
5	B				

1. **γ2 = C**: row 2 allows {B, C, D, E}; column γ allows {A, C}.
2. **α2 = D**: row 2 allows {B, D, E}; column α allows {A, C, D}.
3. **α4 = C**: row 4 allows {B, C, D}; column α allows {A, C}.
4. **ε4 = B**: row 4 allows {B, D}; column ε allows {B, C, E}.
5. **δ2 = B**: in column δ, B fits only in row 2.
6. **β2 = E**: row 2 allows {E}; column β allows {B, C, D, E}.

Exercise 184 | Answer: A

High • 6 steps in this route

	α	β	γ	δ	ε
1	C	A	D		
2			**A**	D	E
3	E	D			C
4	D	B		C	A
5	A				

1. **α1 = C**: in row 1, C fits only in column α.
2. **α4 = D**: in row 4, D fits only in column α.
3. **α3 = E**: in column α, E fits only in row 3.
4. **β3 = D**: row 3 allows {A, B, D}; column β allows {C, D, E}.
5. **δ2 = D**: in row 2, D fits only in column δ.
6. **γ2 = A**: in row 2, A fits only in column γ.

Worked solutions - High

Exercise 185 | Answer: C

High • 7 steps in this route

	α	β	γ	δ	ε
1	C		A	E	D
2	E	A			C
3	A	C	E	D	B
4	D			**C**	
5	B	D	C		

1. **α2 = E**: row 2 allows {B, C, D, E}; column α allows {E}.
2. **ε3 = B**: row 3 allows {B}; column ε allows {A, B, C, D, E}.
3. **ε1 = D**: row 1 allows {B, D}; column ε allows {A, C, D, E}.
4. **ε2 = C**: row 2 allows {B, C, D}; column ε allows {A, C, E}.
5. **β5 = D**: in column β, D fits only in row 5.
6. **γ5 = C**: row 5 allows {A, C, E}; column γ allows {B, C, D}.
7. **δ4 = C**: in row 4, C fits only in column δ.

Exercise 186 | Answer: A

High • 7 steps in this route

	α	β	γ	δ	ε
1	D	B	C		E
2	E			C	**A**
3	B		A	E	
4		E	D	B	C
5			E	D	B

1. **ε1 = E**: in row 1, E fits only in column ε.
2. **δ4 = B**: in column δ, B fits only in row 4.
3. **α1 = D**: in row 1, D fits only in column α.
4. **α3 = B**: in column α, B fits only in row 3.
5. **γ4 = D**: in row 4, D fits only in column γ.
6. **γ3 = A**: row 3 allows {A, C, D}; column γ allows {A, B}.
7. **ε2 = A**: in column ε, A fits only in row 2.

Exercise 187 | Answer: E

High • 7 steps in this route

	α	β	γ	δ	ε
1	C	E		B	A
2		B	C		E
3			B	E	
4	E	D	A	C	
5	B		**E**	A	

1. **γ2 = C**: in row 2, C fits only in column γ.
2. **δ3 = E**: in column δ, E fits only in row 3.
3. **γ4 = A**: in column γ, A fits only in row 4.
4. **α4 = E**: in column α, E fits only in row 4.
5. **β4 = D**: row 4 allows {B, D}; column β allows {A, C, D, E}.
6. **β1 = E**: row 1 allows {D, E}; column β allows {A, C, E}.
7. **γ5 = E**: in row 5, E fits only in column γ.

Exercise 188 | Answer: A

High • 7 steps in this route

	α	β	γ	δ	ε
1		E		D	
2	C		A	B	E
3		B	D		C
4	D		B		**A**
5	B	A	E		D

1. **γ3 = D**: in column γ, D fits only in row 3.
2. **β3 = B**: row 3 allows {A, B, E}; column β allows {B, C, D}.
3. **γ5 = E**: in column γ, E fits only in row 5.
4. **α5 = B**: row 5 allows {B, C}; column α allows {A, B, E}.
5. **δ2 = B**: in column δ, B fits only in row 2.
6. **ε2 = E**: row 2 allows {D, E}; column ε allows {A, B, E}.
7. **ε4 = A**: row 4 allows {A, C, E}; column ε allows {A, B}.

Worked solutions - High

Exercise 189 | Answer: D

High • 7 steps in this route

	α	β	γ	δ	ε
1	C	E		B	
2	E		A		C
3	D	A			B
4		C	B	E	
5	B	**D**		A	E

1. **ε5 = E**: in column ε, E fits only in row 5.
2. **α5 = B**: row 5 allows {B, C, D}; column α allows {A, B, E}.
3. **α2 = E**: in column α, E fits only in row 2.
4. **γ2 = A**: in row 2, A fits only in column γ.
5. **γ4 = B**: in column γ, B fits only in row 4.
6. **β4 = C**: in row 4, C fits only in column β.
7. **β5 = D**: row 5 allows {C, D}; column β allows {B, D}.

Exercise 190 | Answer: E

High • 7 steps in this route

	α	β	γ	δ	ε
1	A	E	B		D
2	C		A		E
3		A			B
4		D	**E**	A	
5	E	C	D		A

1. **ε2 = E**: row 2 allows {A, B, D, E}; column ε allows {C, E}.
2. **β1 = E**: in column β, E fits only in row 1.
3. **β5 = C**: in column β, C fits only in row 5.
4. **γ5 = D**: row 5 allows {B, D}; column γ allows {A, C, D, E}.
5. **γ2 = A**: row 2 allows {A, B, D}; column γ allows {A, C, E}.
6. **δ4 = A**: in row 4, A fits only in column δ.
7. **γ4 = E**: in row 4, E fits only in column γ.

Exercise 191 | Answer: A

High • 7 steps in this route

	α	β	γ	δ	ε
1	**A**		D	E	C
2	B				D
3					B
4	E	C	B	D	
5	D	A	C	B	E

1. **ε2 = D**: in column ε, D fits only in row 2.
2. **γ4 = B**: in column γ, B fits only in row 4.
3. **ε5 = E**: in column ε, E fits only in row 5.
4. **β5 = A**: row 5 allows {A}; column β allows {A, B, C, D, E}.
5. **β4 = C**: row 4 allows {A, C}; column β allows {B, C, D, E}.
6. **ε1 = C**: in column ε, C fits only in row 1.
7. **α1 = A**: row 1 allows {A, B}; column α allows {A, C}.

Exercise 192 | Answer: E

High • 7 steps in this route

	α	β	γ	δ	ε
1		E	C	A	D
2	C			D	A
3	D	A		**E**	
4	E	D	A		
5	A	C		B	

1. **ε1 = D**: in row 1, D fits only in column ε.
2. **γ4 = A**: in row 4, A fits only in column γ.
3. **β4 = D**: in row 4, D fits only in column β.
4. **β5 = C**: row 5 allows {C, D, E}; column β allows {B, C}.
5. **α2 = C**: in row 2, C fits only in column α.
6. **α4 = E**: in column α, E fits only in row 4.
7. **δ3 = E**: in column δ, E fits only in row 3.

Worked solutions - High

Exercise 193 | Answer: B

High • 7 steps in this route

	α	β	γ	δ	ε
1			D		B
2	D	C	E	B	A
3		**B**	C	E	D
4	C		B		
5		E		D	C

1. **α2 = D**: row 2 allows {C, D}; column α allows {A, B, D, E}.
2. **β2 = C**: row 2 allows {C}; column β allows {A, B, C, D}.
3. **γ3 = C**: in row 3, C fits only in column γ.
4. **δ5 = D**: in row 5, D fits only in column δ.
5. **ε5 = C**: in row 5, C fits only in column ε.
6. **ε1 = B**: in column ε, B fits only in row 1.
7. **β3 = B**: in column β, B fits only in row 3.

Exercise 194 | Answer: D

High • 7 steps in this route

	α	β	γ	δ	ε
1	C	B		A	**D**
2	B	C			A
3	E			B	
4	A			D	
5		A	B	C	E

1. **α4 = A**: in column α, A fits only in row 4.
2. **α2 = B**: in column α, B fits only in row 2.
3. **δ3 = B**: in column δ, B fits only in row 3.
4. **δ5 = C**: in column δ, C fits only in row 5.
5. **ε5 = E**: in row 5, E fits only in column ε.
6. **α1 = C**: in column α, C fits only in row 1.
7. **ε1 = D**: row 1 allows {D, E}; column ε allows {B, C, D}.

Exercise 195 | Answer: D

High • 7 steps in this route

	α	β	γ	δ	ε
1	**D**	E		A	
2	A			B	
3	C	B		D	A
4			D	C	B
5	B	C	A		

1. **δ4 = C**: in row 4, C fits only in column δ.
2. **β3 = B**: in column β, B fits only in row 3.
3. **δ3 = D**: in row 3, D fits only in column δ.
4. **δ1 = A**: row 1 allows {A, B, C, D}; column δ allows {A, E}.
5. **γ5 = A**: in row 5, A fits only in column γ.
6. **α5 = B**: in row 5, B fits only in column α.
7. **α1 = D**: row 1 allows {B, C, D}; column α allows {D, E}.

Exercise 196 | Answer: A

High • 8 steps in this route

	α	β	γ	δ	ε
1		E	C		
2	C	A	B		E
3		C	D	A	B
4		D		E	C
5	**A**		E	C	D

1. **γ1 = C**: in row 1, C fits only in column γ.
2. **ε2 = E**: in column ε, E fits only in row 2.
3. **β2 = A**: in row 2, A fits only in column β.
4. **δ4 = E**: in column δ, E fits only in row 4.
5. **β4 = D**: in column β, D fits only in row 4.
6. **γ3 = D**: in column γ, D fits only in row 3.
7. **γ5 = E**: in column γ, E fits only in row 5.
8. **α5 = A**: in row 5, A fits only in column α.

Worked solutions - High

Exercise 197 | Answer: A

High • 8 steps in this route

	α	β	γ	δ	ε
1				C	A
2			A	B	E
3	C	**A**	E	D	
4	B		C		D
5	A	B	D	E	

1. **ε4 = D**: row 4 allows {A, D, E}; column ε allows {B, C, D}.
2. **γ2 = A**: in column γ, A fits only in row 2.
3. **δ2 = B**: in row 2, B fits only in column δ.
4. **α5 = A**: in column α, A fits only in row 5.
5. **δ5 = E**: in row 5, E fits only in column δ.
6. **δ1 = C**: in column δ, C fits only in row 1.
7. **δ3 = D**: in column δ, D fits only in row 3.
8. **β3 = A**: row 3 allows {A, B}; column β allows {A, C, D, E}.

Exercise 198 | Answer: E

High • 8 steps in this route

	α	β	γ	δ	ε
1	C	E		B	A
2		B	A		
3	B	D	C	A	**E**
4			B	E	
5	D	A		C	

1. **α3 = B**: in column α, B fits only in row 3.
2. **δ3 = A**: in row 3, A fits only in column δ.
3. **δ5 = C**: row 5 allows {A, B, C, E}; column δ allows {C, D}.
4. **β5 = A**: in row 5, A fits only in column β.
5. **β1 = E**: in column β, E fits only in row 1.
6. **α1 = C**: row 1 allows {C, D}; column α allows {A, C, E}.
7. **γ3 = C**: in column γ, C fits only in row 3.
8. **ε3 = E**: row 3 allows {E}; column ε allows {B, C, D, E}.

Exercise 199 | Answer: D

High • 8 steps in this route

	α	β	γ	δ	ε
1	D		A	C	
2	A	C			D
3			D		C
4	B		C	E	A
5	C		B	**D**	E

1. **α5 = C**: in row 5, C fits only in column α.
2. **γ4 = C**: in column γ, C fits only in row 4.
3. **γ1 = A**: in column γ, A fits only in row 1.
4. **ε3 = C**: in row 3, C fits only in column ε.
5. **α1 = D**: in column α, D fits only in row 1.
6. **ε4 = A**: in column ε, A fits only in row 4.
7. **ε2 = D**: in column ε, D fits only in row 2.
8. **δ5 = D**: in column δ, D fits only in row 5.

Exercise 200 | Answer: D

High • 8 steps in this route

	α	β	γ	δ	ε
1	A	D			E
2	B	C	E		
3		B	D		A
4	E		C	**D**	B
5		E	A	B	

1. **β5 = E**: in row 5, E fits only in column β.
2. **γ2 = E**: in column γ, E fits only in row 2.
3. **β3 = B**: row 3 allows {B, C, E}; column β allows {A, B, D}.
4. **ε4 = B**: in row 4, B fits only in column ε.
5. **α2 = B**: in row 2, B fits only in column α.
6. **α1 = A**: in column α, A fits only in row 1.
7. **β1 = D**: row 1 allows {B, C, D}; column β allows {A, D}.
8. **δ4 = D**: in row 4, D fits only in column δ.

Worked solutions - High

Exercise 201 | Answer: C

High • 8 steps in this route

	α	β	γ	δ	ε
1	C		B		
2	E	B	A		
3	B	D		A	E
4		C	D	E	
5	D		E	B	**C**

1. **α3 = B**: row 3 allows {B, C, D}; column α allows {A, B, E}.
2. **β3 = D**: row 3 allows {C, D}; column β allows {A, B, D, E}.
3. **γ2 = A**: in column γ, A fits only in row 2.
4. **α2 = E**: row 2 allows {B, C, D, E}; column α allows {A, E}.
5. **β2 = B**: row 2 allows {B, C, D}; column β allows {A, B, E}.
6. **δ4 = E**: in row 4, E fits only in column δ.
7. **δ5 = B**: in column δ, B fits only in row 5.
8. **ε5 = C**: in row 5, C fits only in column ε.

Exercise 202 | Answer: A

High • 9 steps in this route

	α	β	γ	δ	ε
1		D	C	E	
2	D	C	E	A	
3		A	D	B	C
4	**A**		B		D
5	C	B			E

1. **β1 = D**: row 1 allows {A, B, C, D}; column β allows {D, E}.
2. **ε5 = E**: in row 5, E fits only in column ε.
3. **ε3 = C**: row 3 allows {C, D, E}; column ε allows {A, B, C}.
4. **γ3 = D**: row 3 allows {D, E}; column γ allows {A, B, C, D}.
5. **γ1 = C**: in row 1, C fits only in column γ.
6. **α2 = D**: in column α, D fits only in row 2.
7. **δ2 = A**: row 2 allows {A, B}; column δ allows {A, C, D}.
8. **γ4 = B**: in column γ, B fits only in row 4.
9. **α4 = A**: in row 4, A fits only in column α.

Exercise 203 | Answer: B

High • 9 steps in this route

	α	β	γ	δ	ε
1		C		D	E
2	C	D	B	E	A
3			C	B	
4	D		E		B
5		**B**	D	A	C

1. **ε4 = B**: row 4 allows {A, B, C, E}; column ε allows {B, D}.
2. **γ5 = D**: in row 5, D fits only in column γ.
3. **δ1 = D**: in row 1, D fits only in column δ.
4. **γ4 = E**: in column γ, E fits only in row 4.
5. **δ3 = B**: in column δ, B fits only in row 3.
6. **δ2 = E**: in column δ, E fits only in row 2.
7. **α2 = C**: row 2 allows {C}; column α allows {A, B, C, E}.
8. **β1 = C**: in row 1, C fits only in column β.
9. **β5 = B**: in column β, B fits only in row 5.

Worked solutions - High

Exercise 204 | Answer: E

High • 10 steps in this route

	α	β	γ	δ	ε
1	E	A		B	
2				**E**	C
3	B	E	D	C	A
4	D	C		A	B
5		B	A	D	E

1. **α3 = B**: row 3 allows {B, E}; column α allows {A, B, C, D}.
2. **β1 = A**: in row 1, A fits only in column β.
3. **β3 = E**: row 3 allows {E}; column β allows {B, C, D, E}.
4. **ε5 = E**: in row 5, E fits only in column ε.
5. **β5 = B**: in row 5, B fits only in column β.
6. **ε4 = B**: in column ε, B fits only in row 4.
7. **β4 = C**: in column β, C fits only in row 4.
8. **α4 = D**: in row 4, D fits only in column α.
9. **δ4 = A**: in row 4, A fits only in column δ.
10. **δ2 = E**: row 2 allows {A, B, D, E}; column δ allows {E}.

Exercise 205 | Answer: C

High • 10 steps in this route

	α	β	γ	δ	ε
1	D	B	**C**		E
2	C	D		E	B
3	A	E		B	C
4		C	E	D	A
5		A	B		

1. **β2 = D**: row 2 allows {A, D}; column β allows {B, C, D}.
2. **ε1 = E**: in row 1, E fits only in column ε.
3. **ε4 = A**: in column ε, A fits only in row 4.
4. **α3 = A**: in column α, A fits only in row 3.
5. **δ4 = D**: in row 4, D fits only in column δ.
6. **δ3 = B**: row 3 allows {B, D}; column δ allows {A, B, C}.
7. **β4 = C**: in row 4, C fits only in column β.
8. **β1 = B**: row 1 allows {A, B, C}; column β allows {B}.
9. **γ5 = B**: in column γ, B fits only in row 5.
10. **γ1 = C**: in column γ, C fits only in row 1.

Exercise 206 | Answer: A

High • 6 steps in this route

	α	β	γ	δ	ε
1	A				E
2	E		B	A	D
3	B				C
4		E	**A**		B
5		B	E	C	

1. **ε1 = E**: in column ε, E fits only in row 1.
2. **ε4 = B**: in column ε, B fits only in row 4.
3. **α3 = B**: in column α, B fits only in row 3.
4. **α2 = E**: in column α, E fits only in row 2.
5. **δ2 = A**: row 2 allows {A, C}; column δ allows {A, B, D, E}.
6. **γ4 = A**: in row 4, A fits only in column γ.

Exercise 207 | Answer: B

High • 7 steps in this route

	α	β	γ	δ	ε
1	C			**B**	A
2	A	C		E	D
3	B	D	E		
4	D				
5		A	C	D	B

1. **α3 = B**: row 3 allows {A, B, C, D}; column α allows {B, E}.
2. **γ5 = C**: in row 5, C fits only in column γ.
3. **β5 = A**: in row 5, A fits only in column β.
4. **β3 = D**: row 3 allows {A, C, D}; column β allows {B, D, E}.
5. **ε2 = D**: in column ε, D fits only in row 2.
6. **δ2 = E**: in row 2, E fits only in column δ.
7. **δ1 = B**: row 1 allows {B, D, E}; column δ allows {A, B, C}.

Worked solutions - High

Exercise 208 | Answer: C

High • 8 steps in this route

	α	β	γ	δ	ε
1	D	C	A		B
2	B	A		D	E
3	A	B			
4	E	D	B	A	**C**
5	C			B	

1. **α2 = B**: in row 2, B fits only in column α.
2. **β2 = A**: in row 2, A fits only in column β.
3. **α5 = C**: row 5 allows {A, B, C, D, E}; column α allows {C}.
4. **β1 = C**: in column β, C fits only in row 1.
5. **ε1 = B**: row 1 allows {B, E}; column ε allows {A, B, C, D}.
6. **δ5 = B**: in row 5, B fits only in column δ.
7. **δ4 = A**: in column δ, A fits only in row 4.
8. **ε4 = C**: row 4 allows {C}; column ε allows {A, C, D}.

Exercise 209 | Answer: A

High • 8 steps in this route

	α	β	γ	δ	ε
1		A	B		C
2			E	B	D
3		B		**A**	E
4	E	D		C	
5	B	E	C		A

1. **ε2 = D**: row 2 allows {A, C, D, E}; column ε allows {B, D}.
2. **α5 = B**: in row 5, B fits only in column α.
3. **γ1 = B**: in row 1, B fits only in column γ.
4. **γ2 = E**: in column γ, E fits only in row 2.
5. **β5 = E**: in column β, E fits only in row 5.
6. **β4 = D**: in column β, D fits only in row 4.
7. **δ4 = C**: in row 4, C fits only in column δ.
8. **δ3 = A**: in column δ, A fits only in row 3.

Exercise 210 | Answer: E

High • 8 steps in this route

	α	β	γ	δ	ε
1		B		C	D
2	B	A	E		C
3		D	B		E
4	D	E	C		A
5			D	**E**	

1. **α4 = D**: row 4 allows {B, D}; column α allows {A, C, D, E}.
2. **ε1 = D**: in row 1, D fits only in column ε.
3. **ε2 = C**: row 2 allows {A, C, D}; column ε allows {B, C, E}.
4. **β2 = A**: row 2 allows {A, D}; column β allows {A, B, C}.
5. **β1 = B**: row 1 allows {A, B, E}; column β allows {B, C}.
6. **γ3 = B**: in column γ, B fits only in row 3.
7. **ε3 = E**: row 3 allows {A, C, E}; column ε allows {B, E}.
8. **δ5 = E**: in column δ, E fits only in row 5.